

BFS Traversal

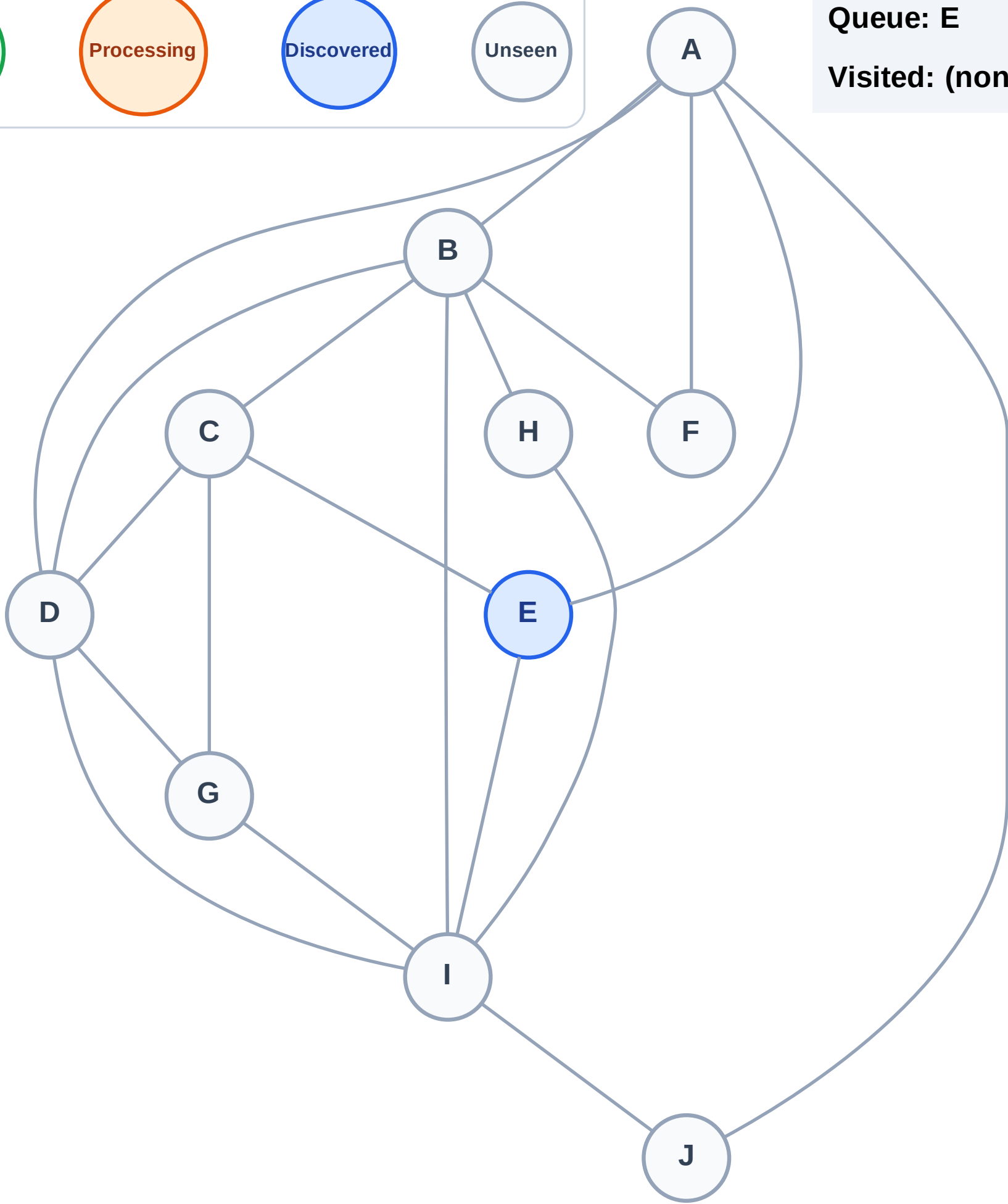
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

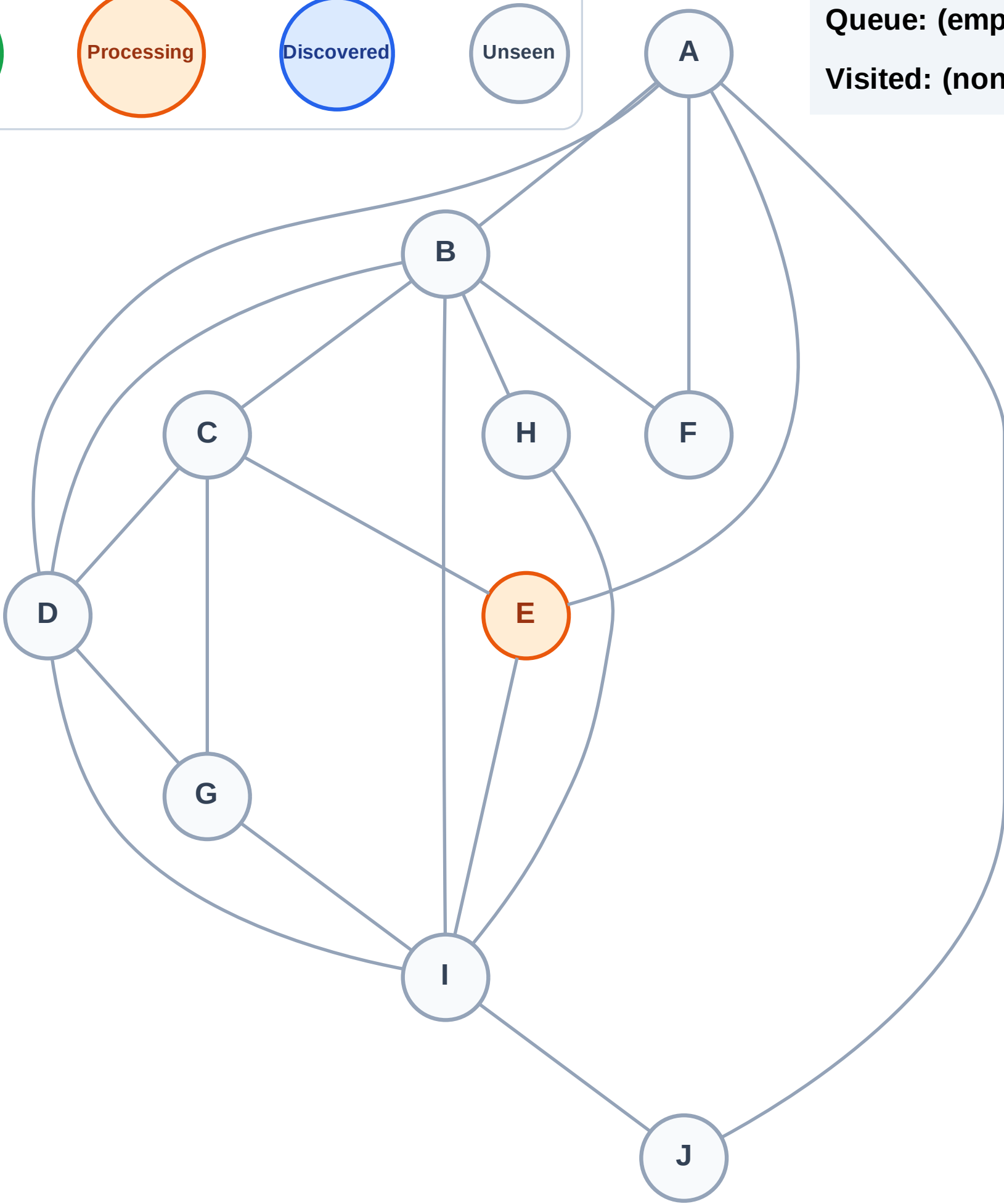
Step 2: Process node E

Legend



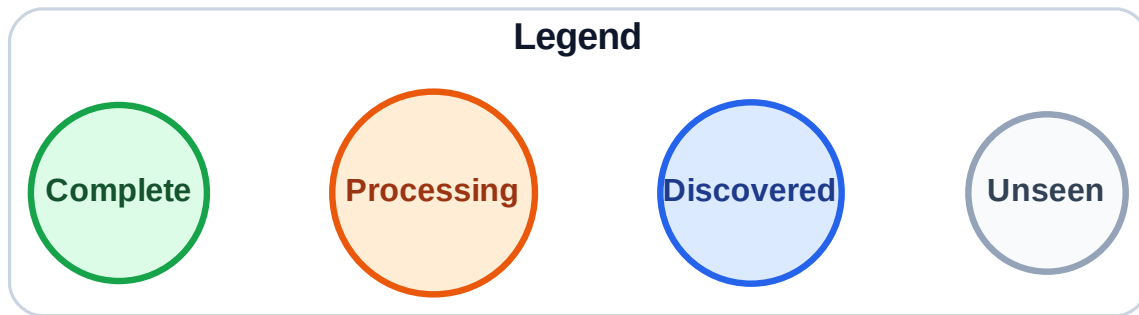
Queue: (empty)

Visited: (none)

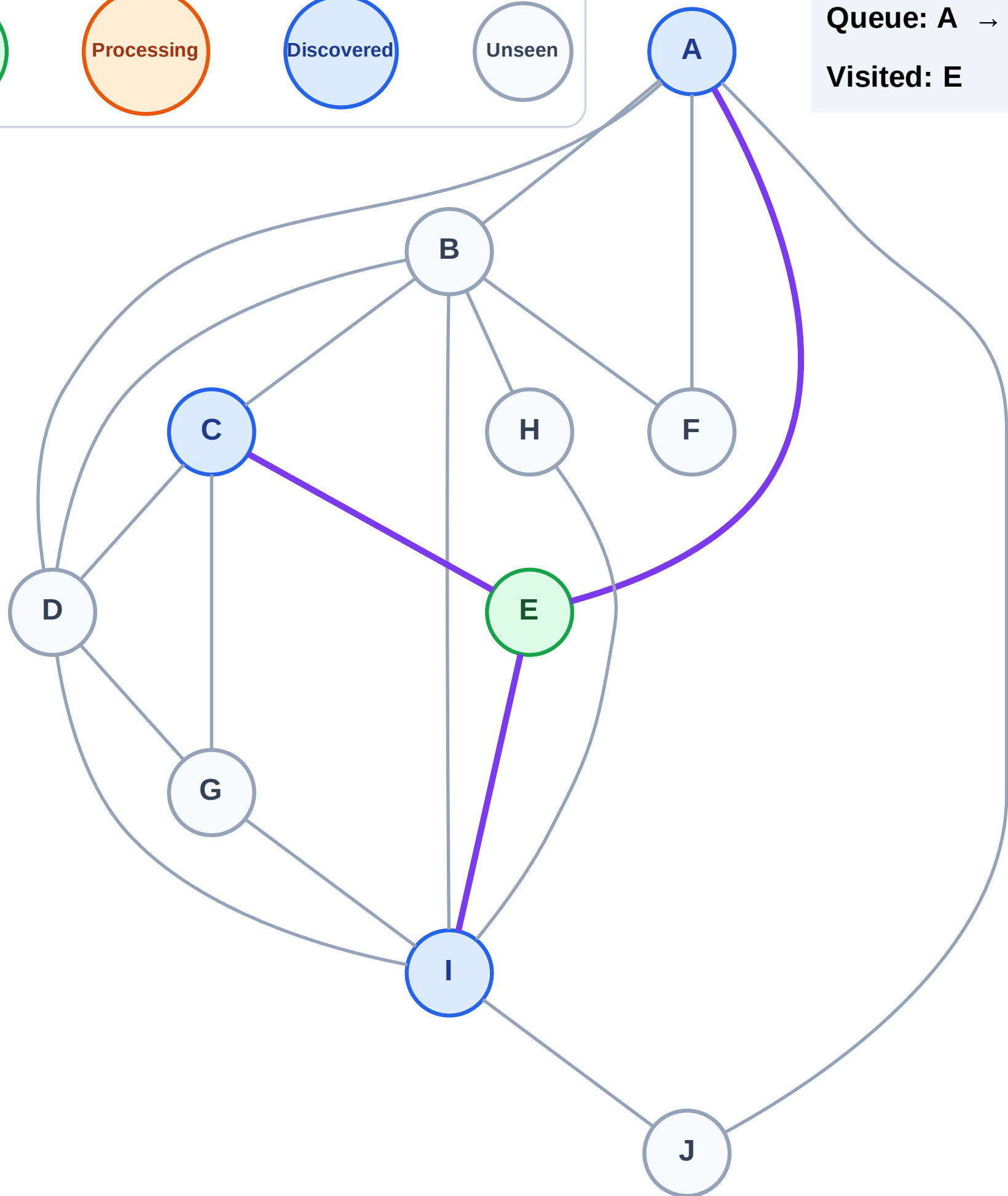


BFS Traversal

Step 3: Discover A, C, I from E



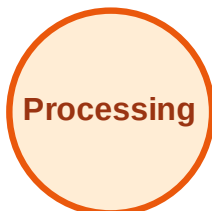
Queue: A → C → I
Visited: E



BFS Traversal

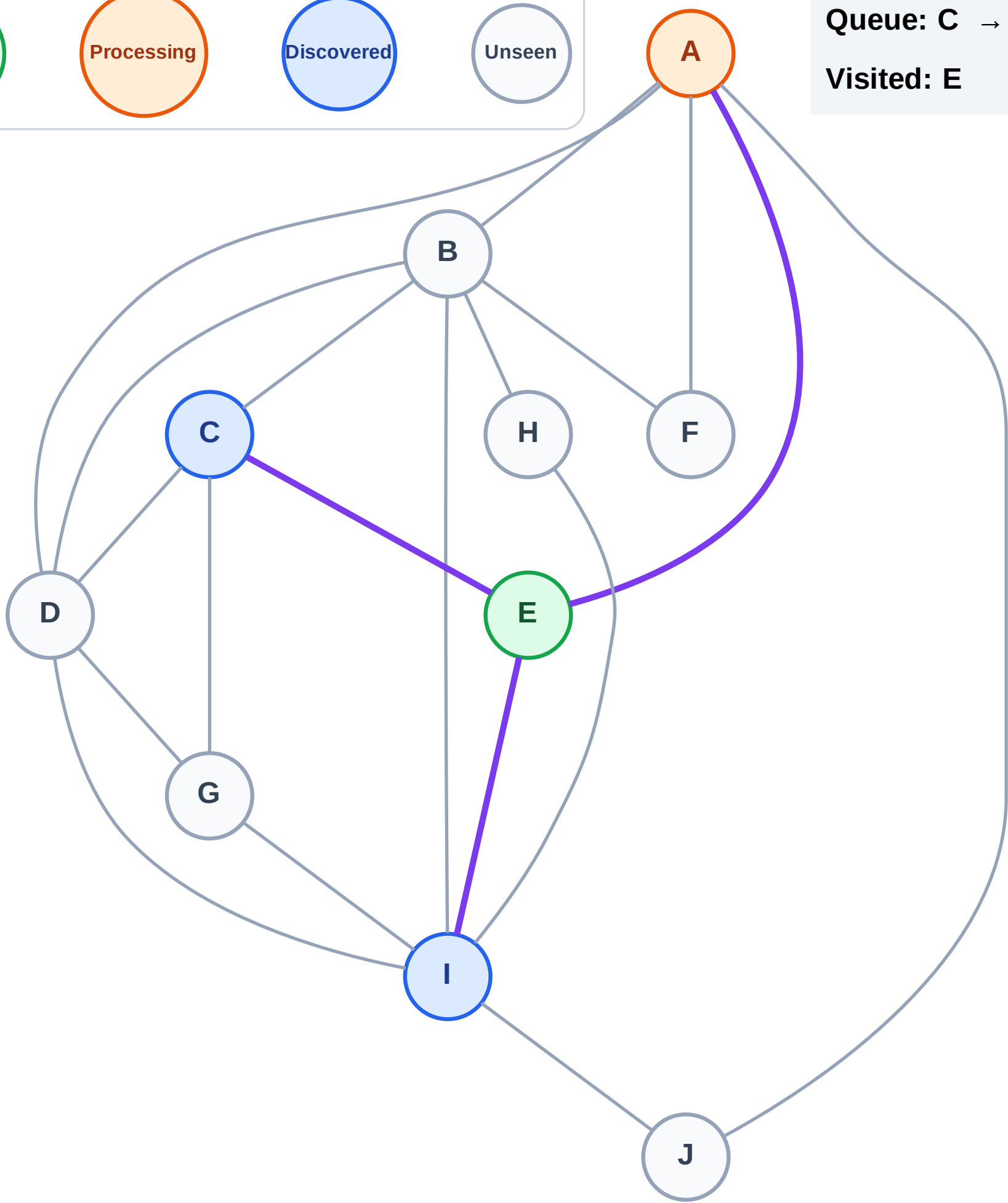
Step 4: Process node A

Legend



Queue: C → I

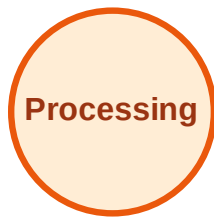
Visited: E



BFS Traversal

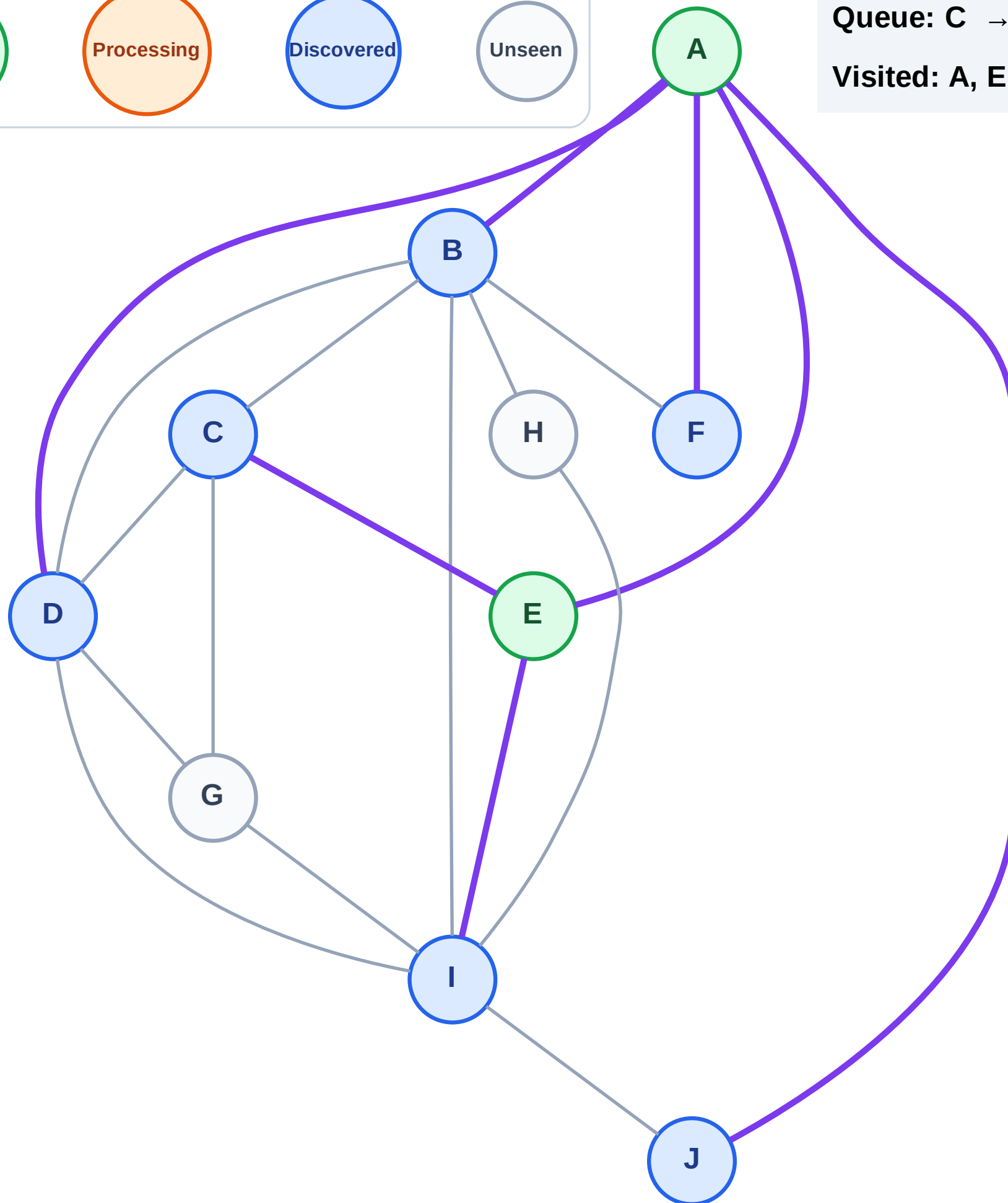
Step 5: Discover B, D, F, J from A

Legend



Queue: C → I → B → D → F → J

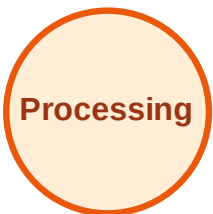
Visited: A, E



BFS Traversal

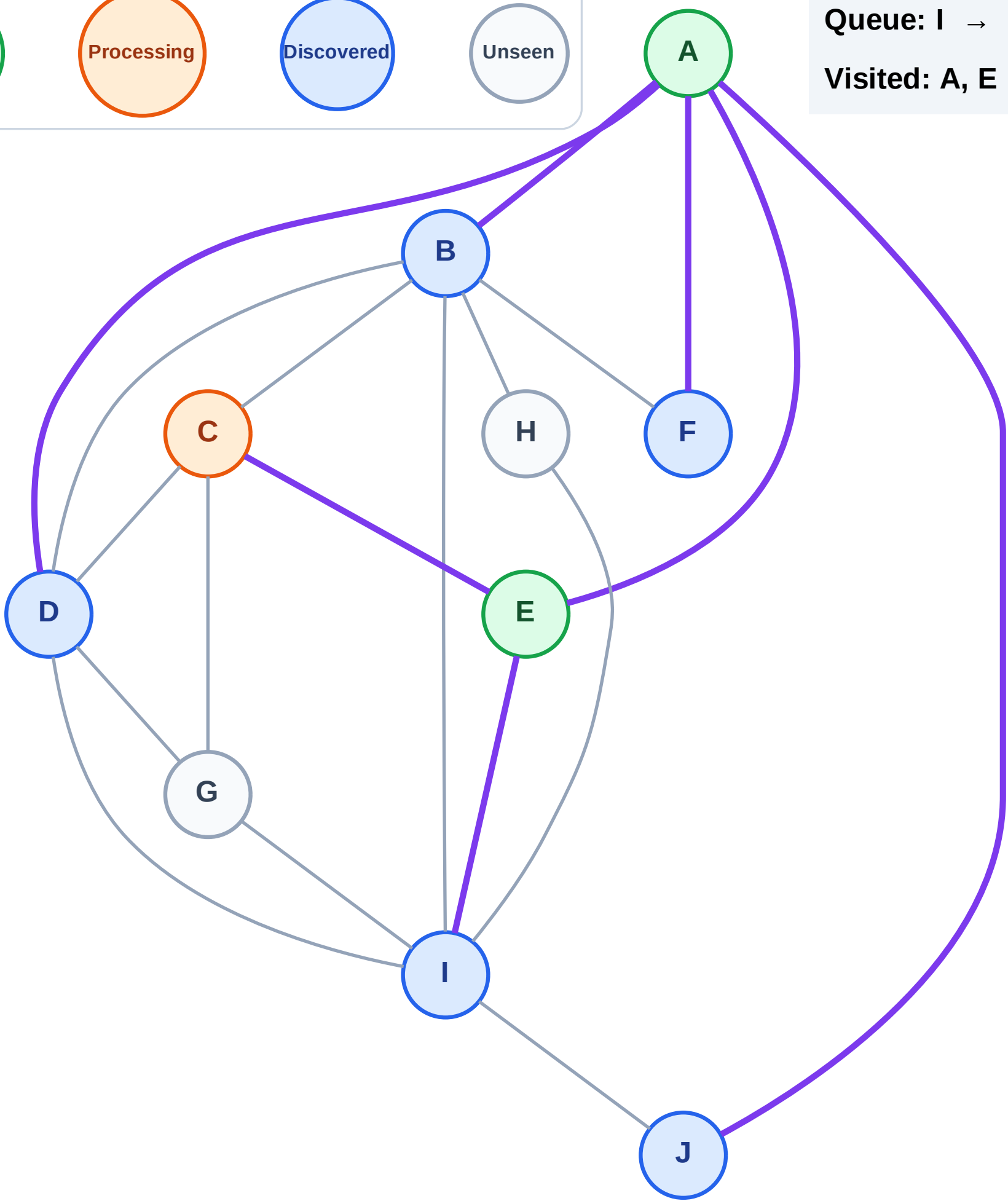
Step 6: Process node C

Legend



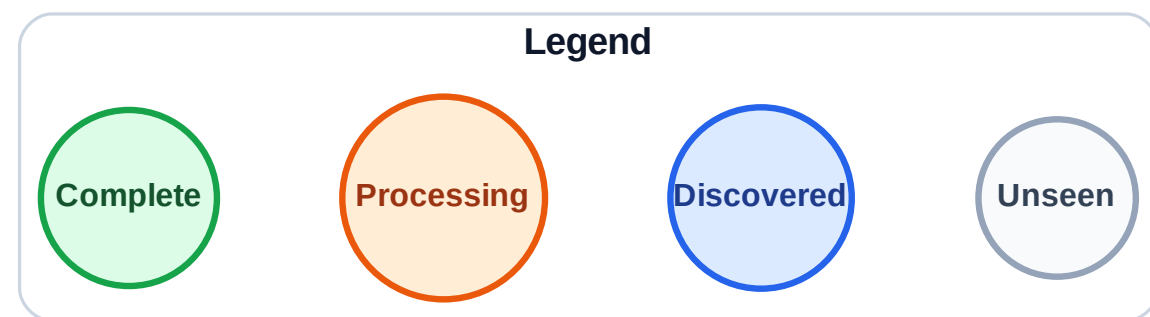
Queue: I → B → D → F → J

Visited: A, E

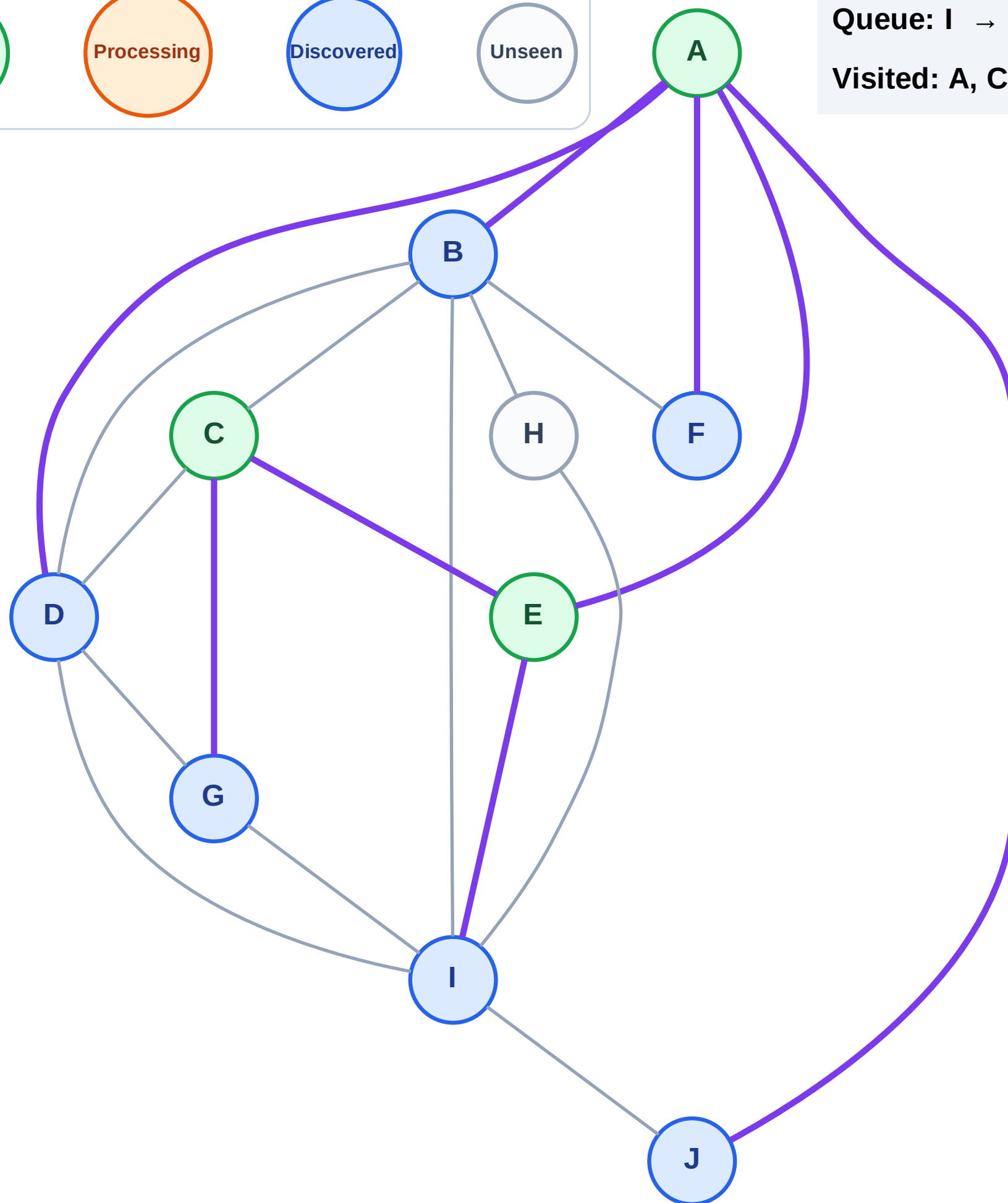


BFS Traversal

Step 7: Discover G from C



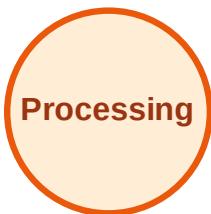
Queue: I → B → D → F → J → G
Visited: A, C, E



BFS Traversal

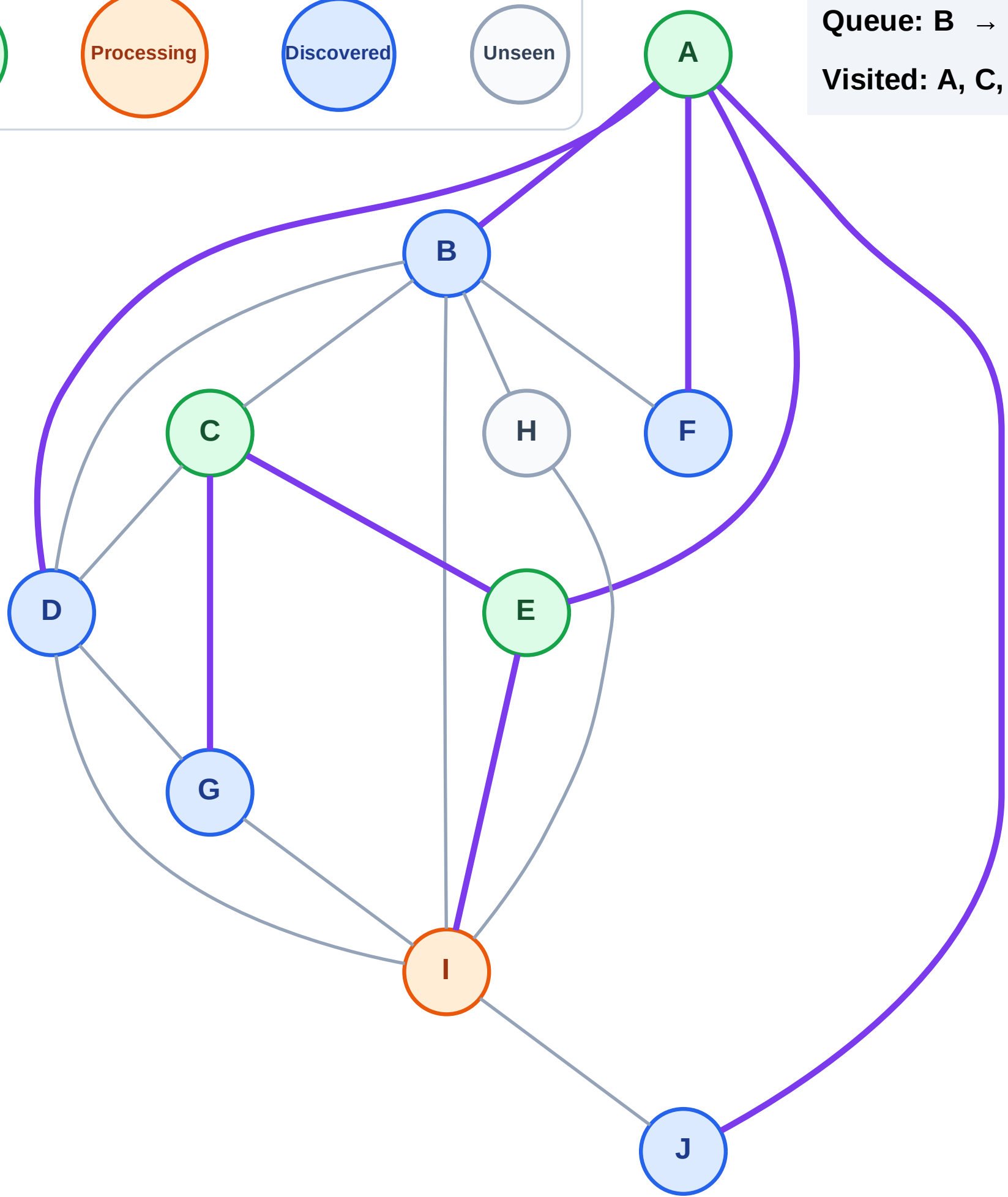
Step 8: Process node I

Legend



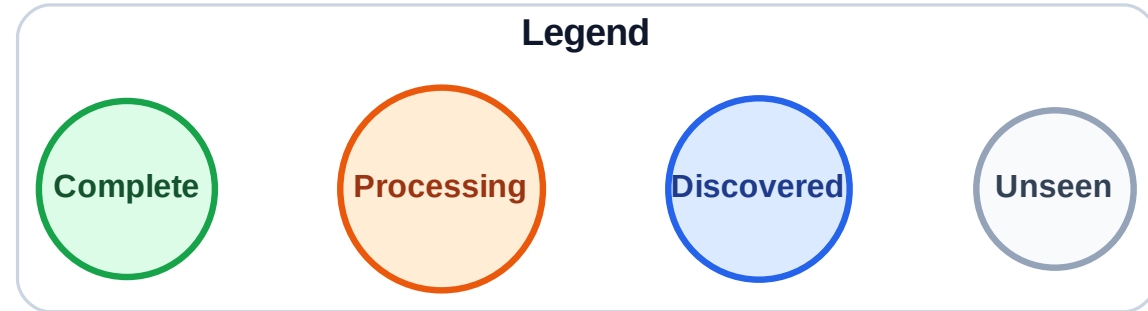
Queue: B → D → F → J → G

Visited: A, C, E



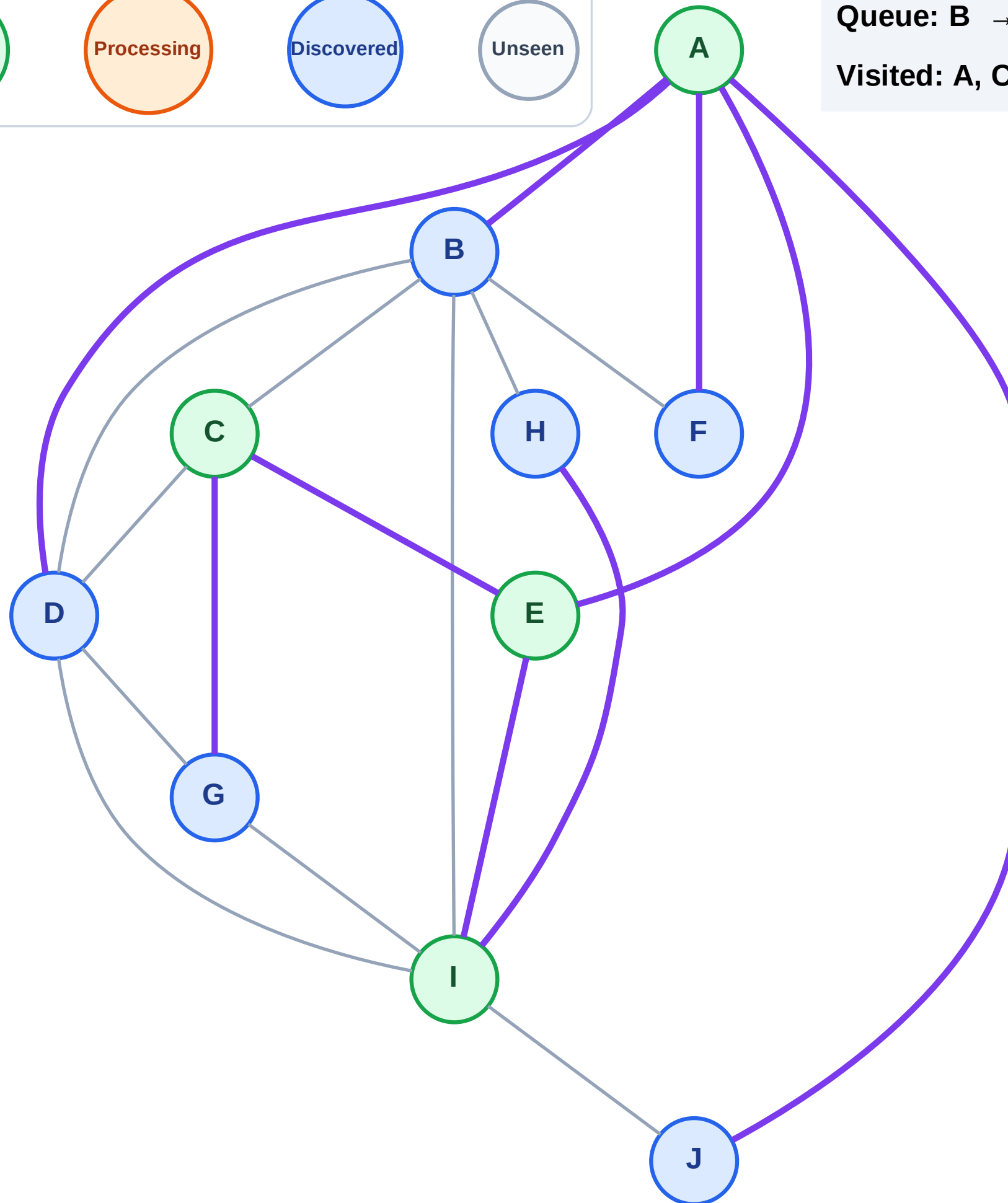
BFS Traversal

Step 9: Discover H from I



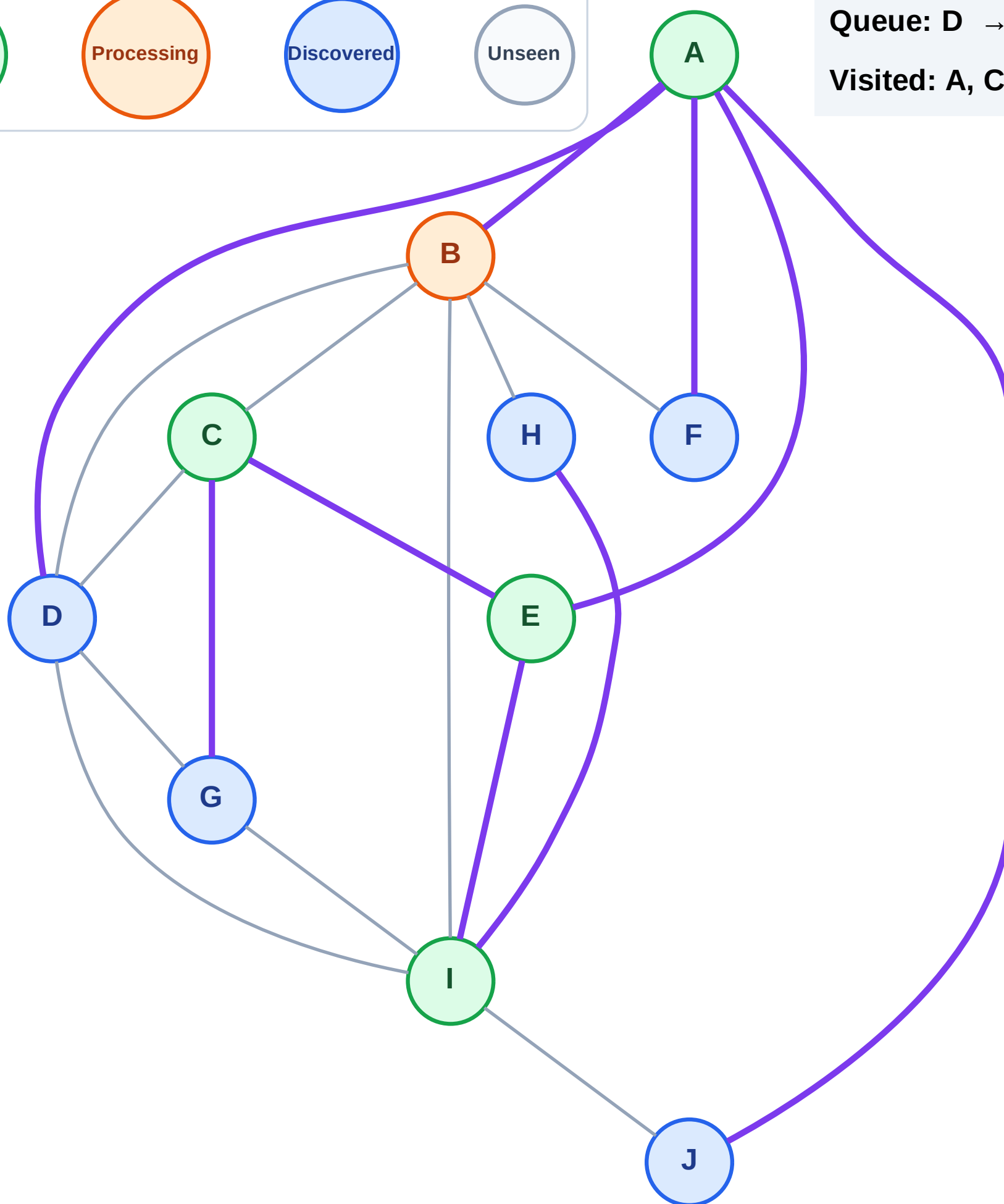
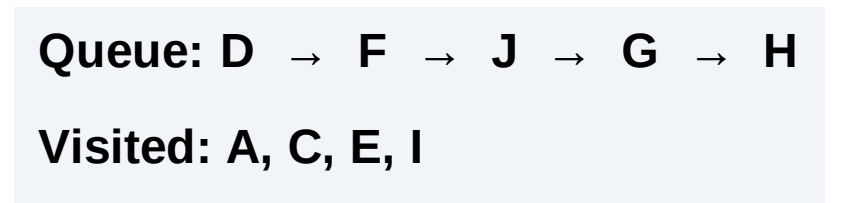
Queue: B → D → F → J → G → H

Visited: A, C, E, I



Step 10: Process node B

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BFS Traversal

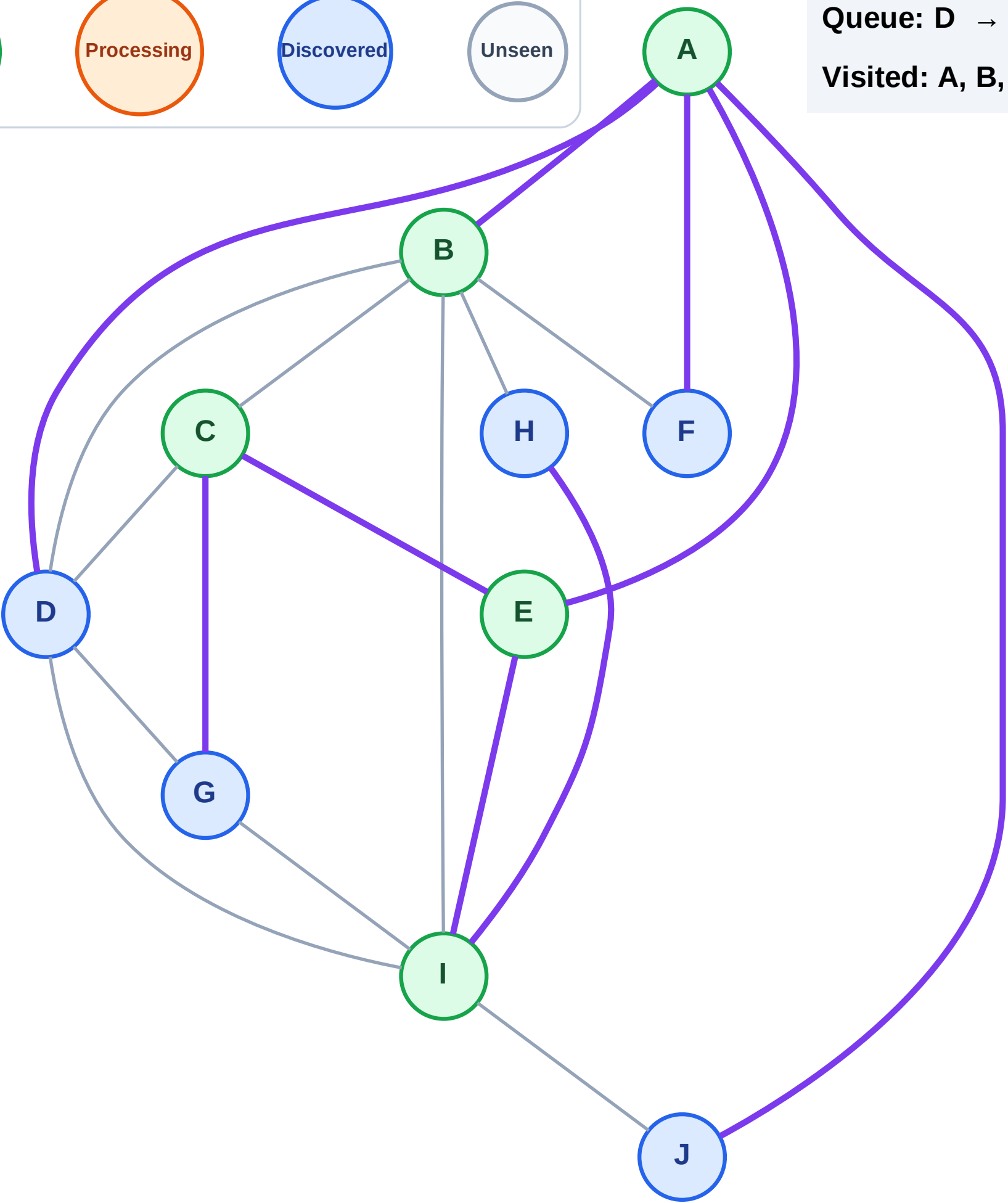
Step 11: No new neighbors from B

Legend



Queue: D → F → J → G → H

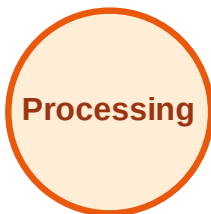
Visited: A, B, C, E, I



BFS Traversal

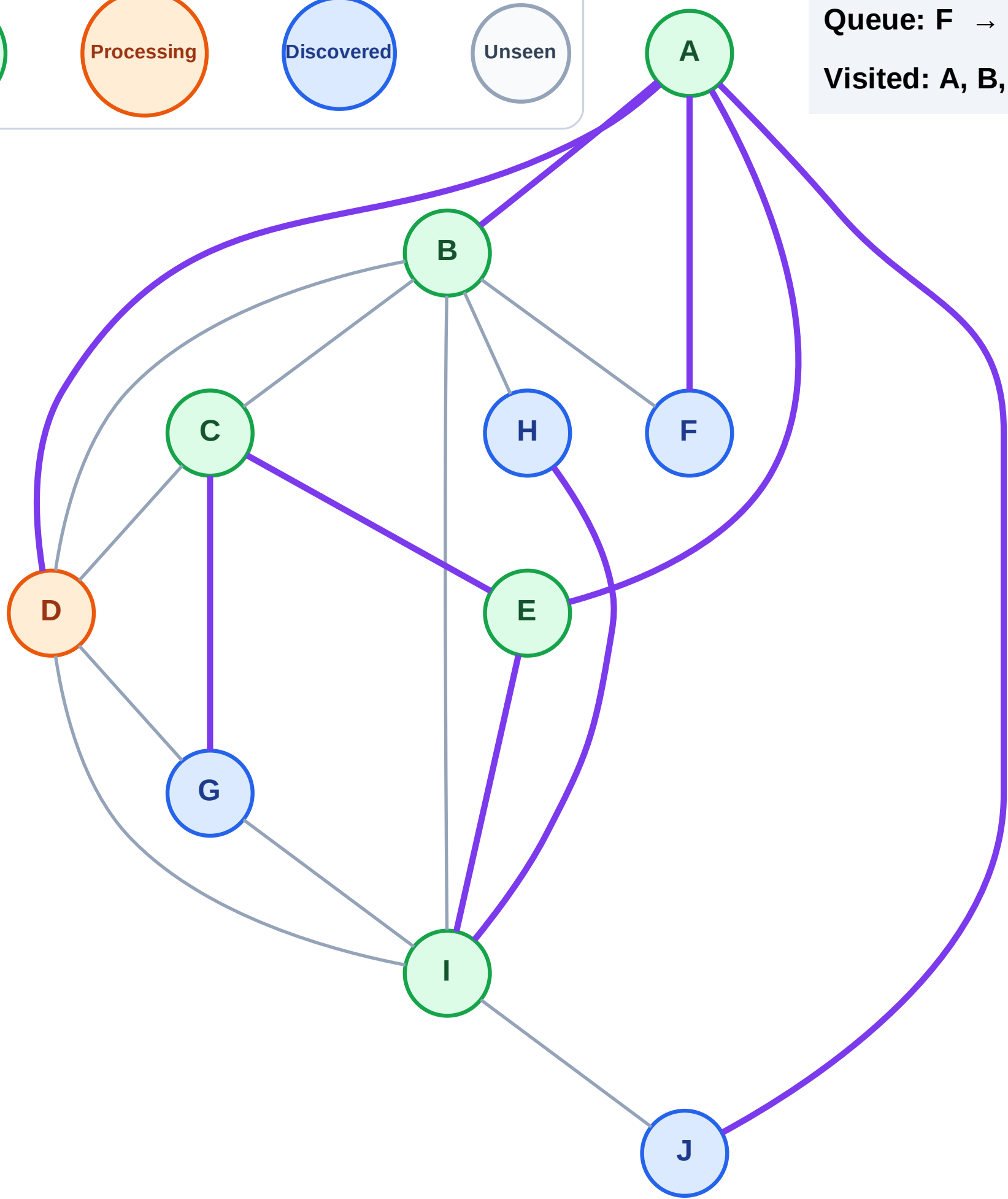
Step 12: Process node D

Legend



Queue: F → J → G → H

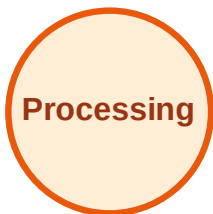
Visited: A, B, C, E, I



BFS Traversal

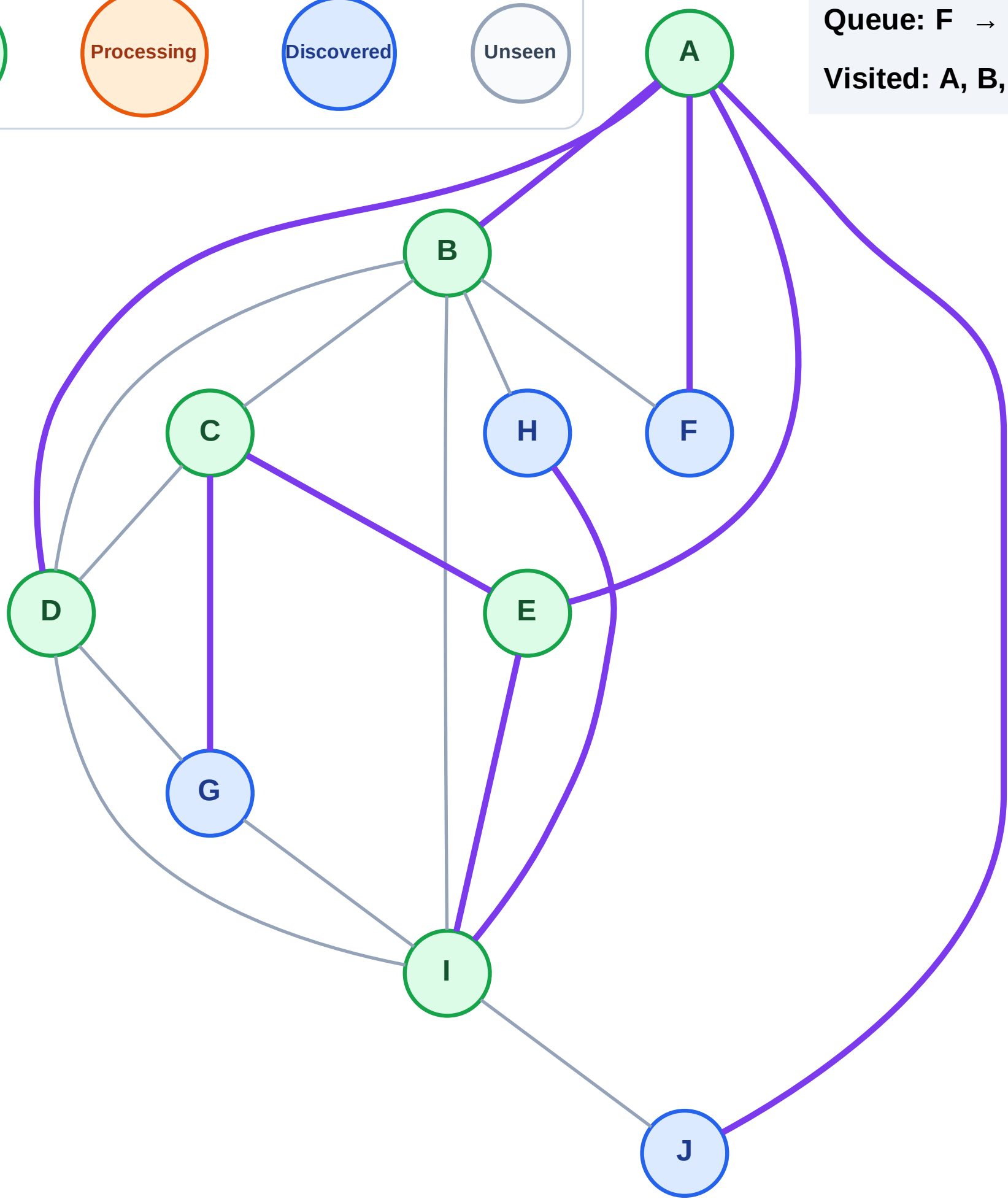
Step 13: No new neighbors from D

Legend



Queue: F → J → G → H

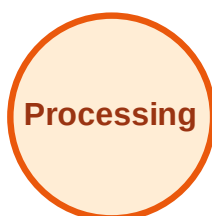
Visited: A, B, C, D, E, I



BFS Traversal

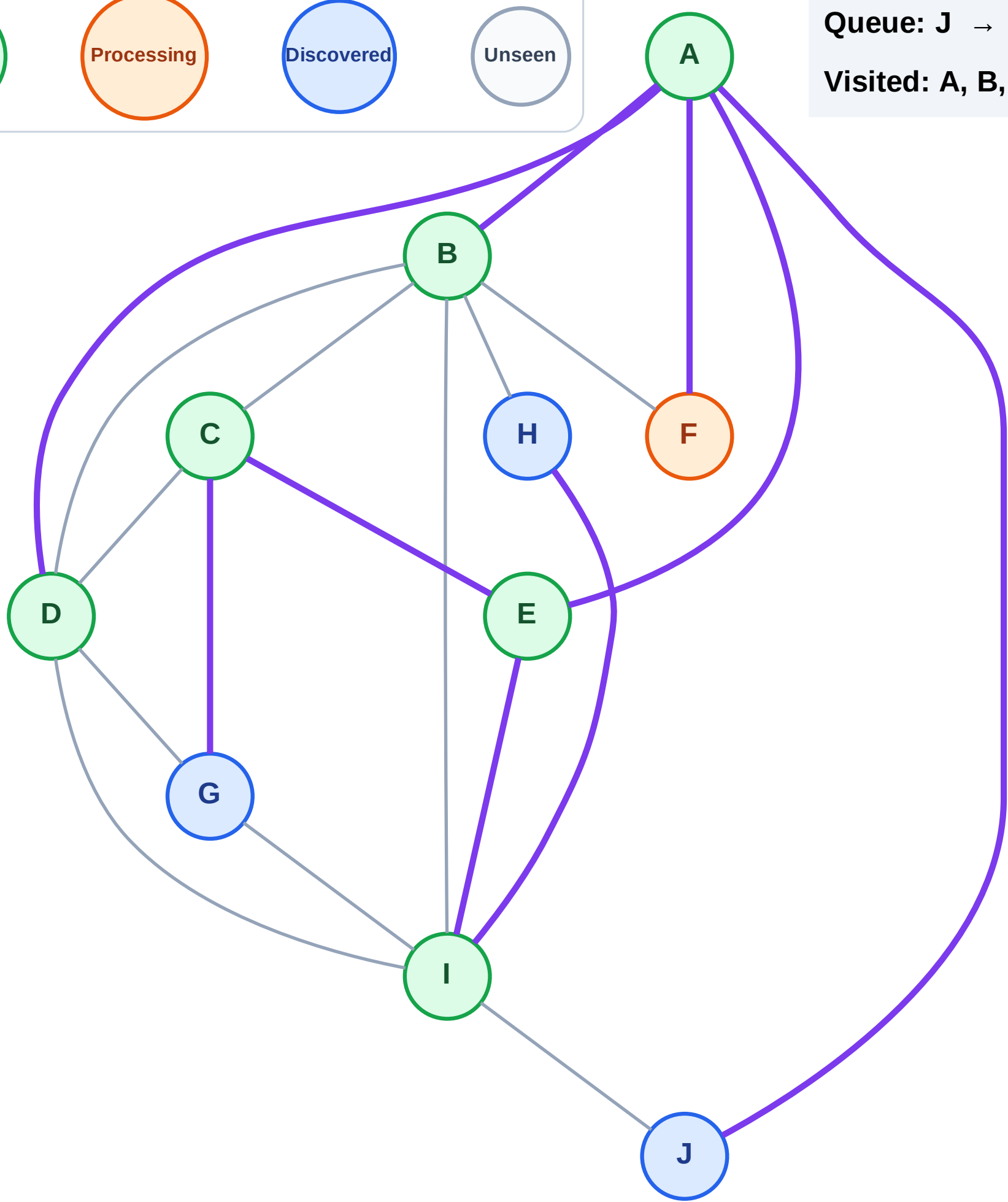
Step 14: Process node F

Legend



Queue: J → G → H

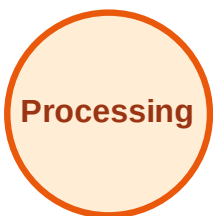
Visited: A, B, C, D, E, I



BFS Traversal

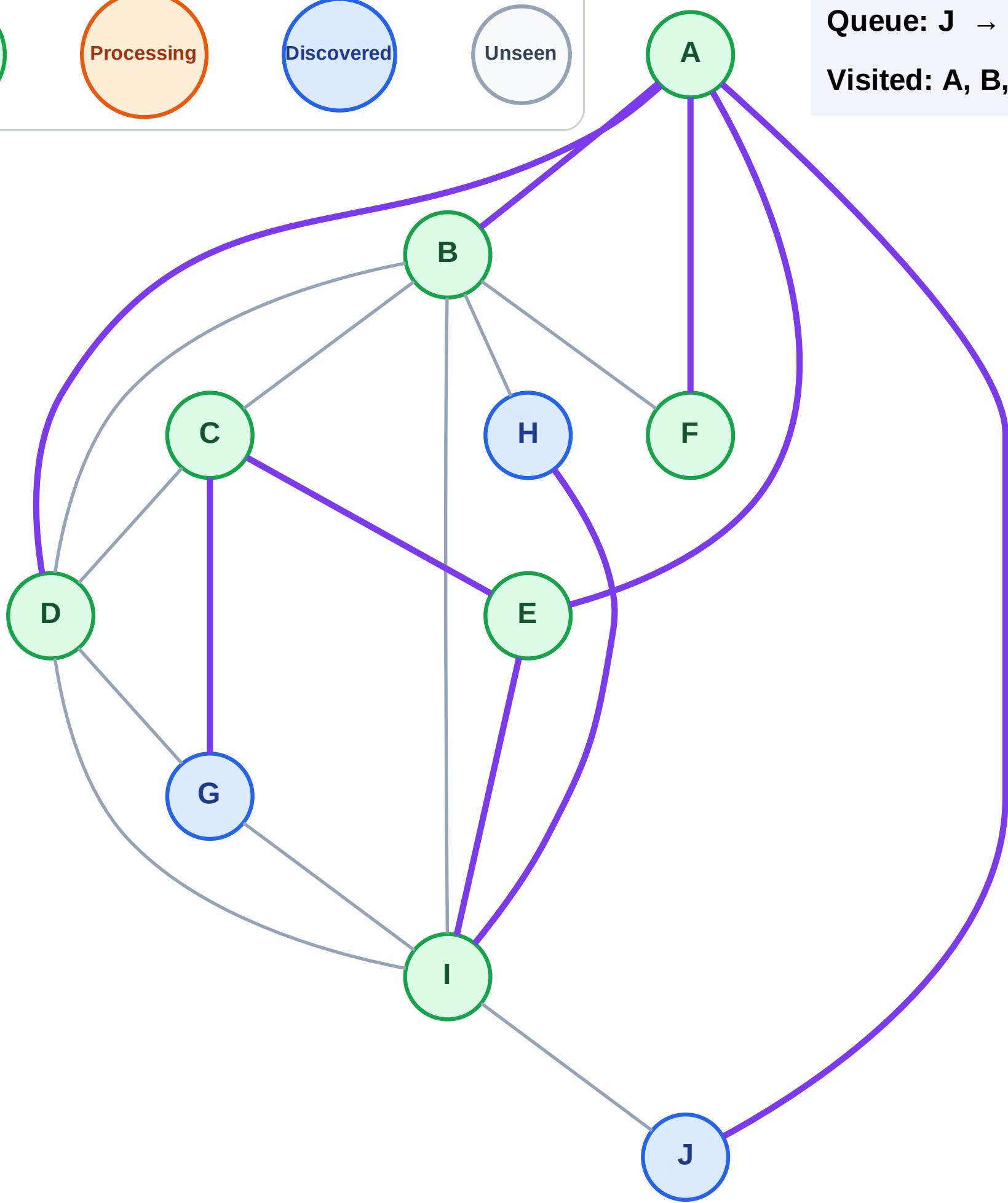
Step 15: No new neighbors from F

Legend



Queue: J → G → H

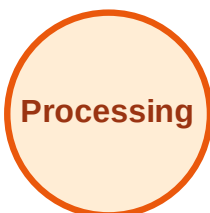
Visited: A, B, C, D, E, F, I



BFS Traversal

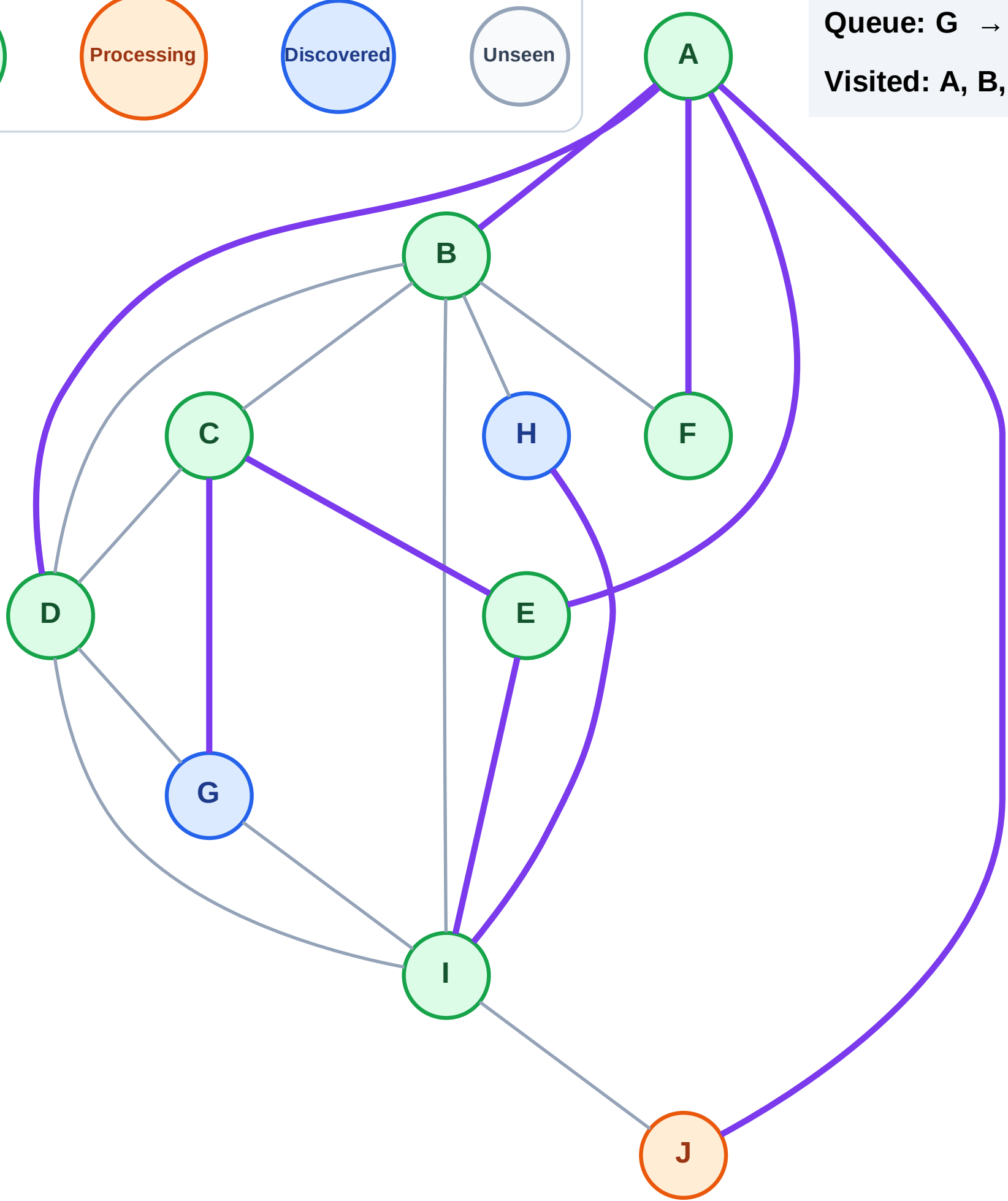
Step 16: Process node J

Legend



Queue: G → H

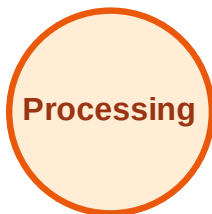
Visited: A, B, C, D, E, F, I



BFS Traversal

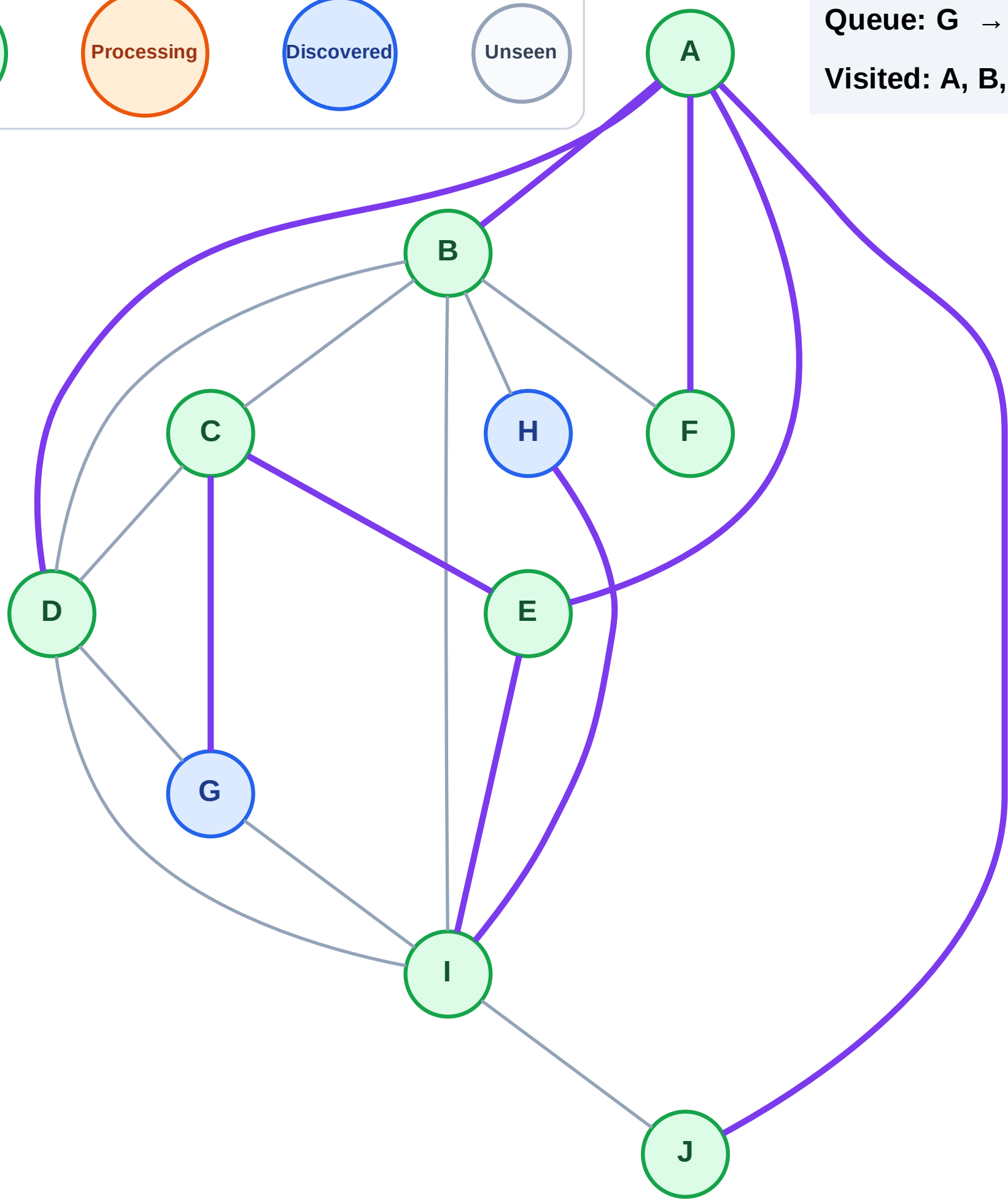
Step 17: No new neighbors from J

Legend



Queue: G → H

Visited: A, B, C, D, E, F, I, J



BFS Traversal

Step 18: Process node G

Legend

Complete

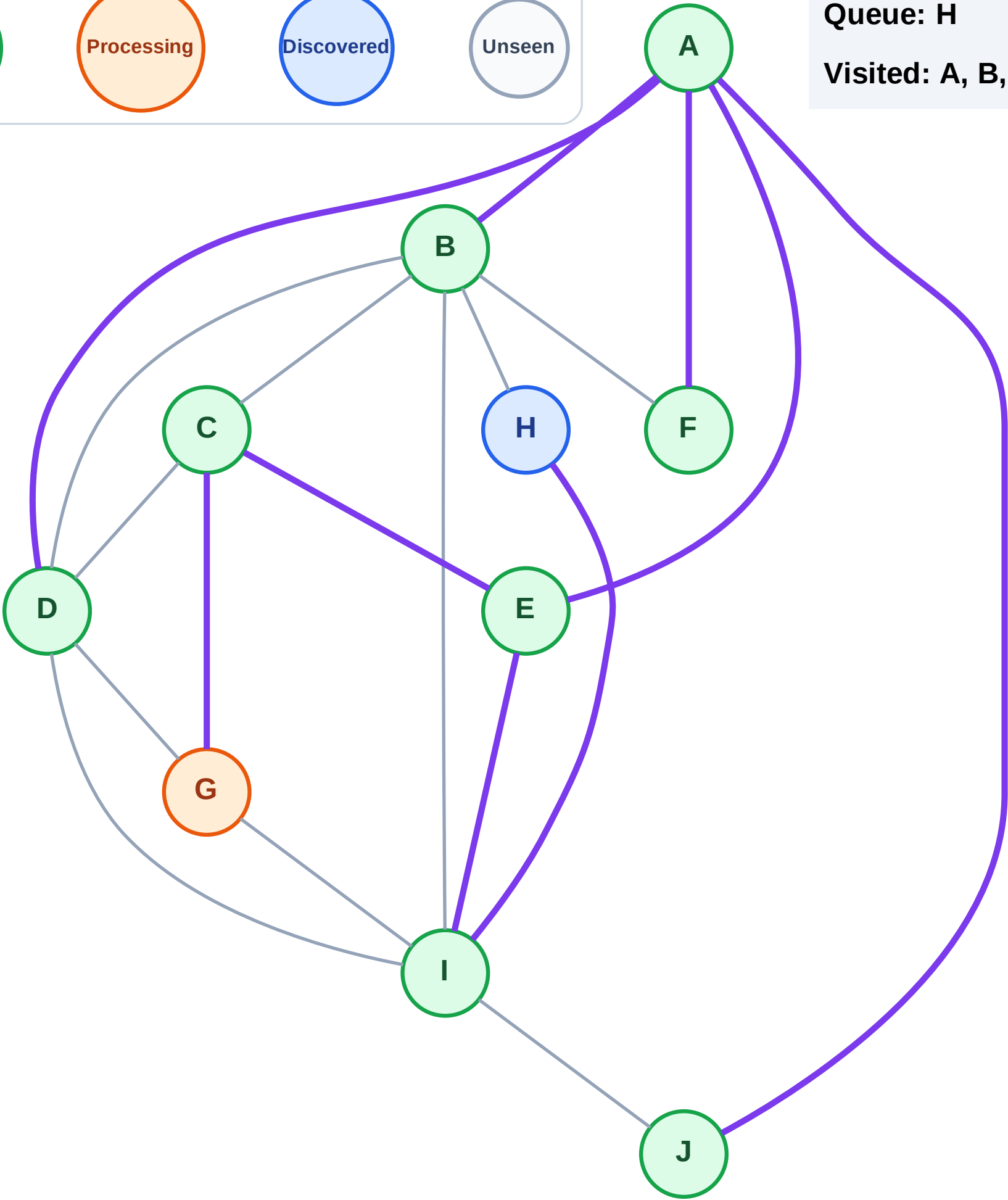
Processing

Discovered

Unseen

Queue: H

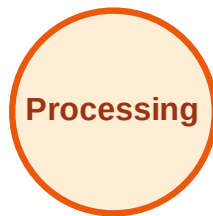
Visited: A, B, C, D, E, F, I, J



BFS Traversal

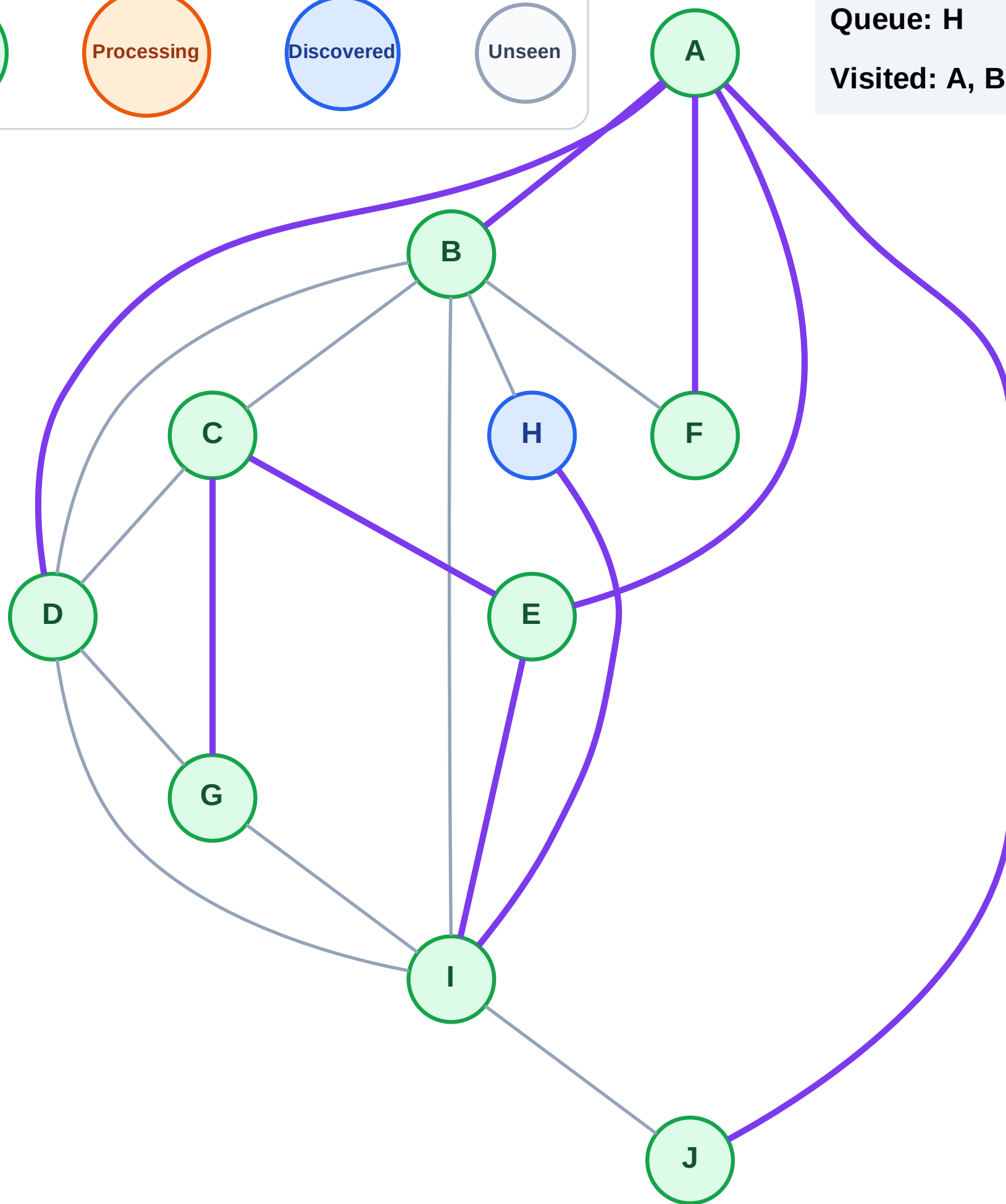
Step 19: No new neighbors from G

Legend



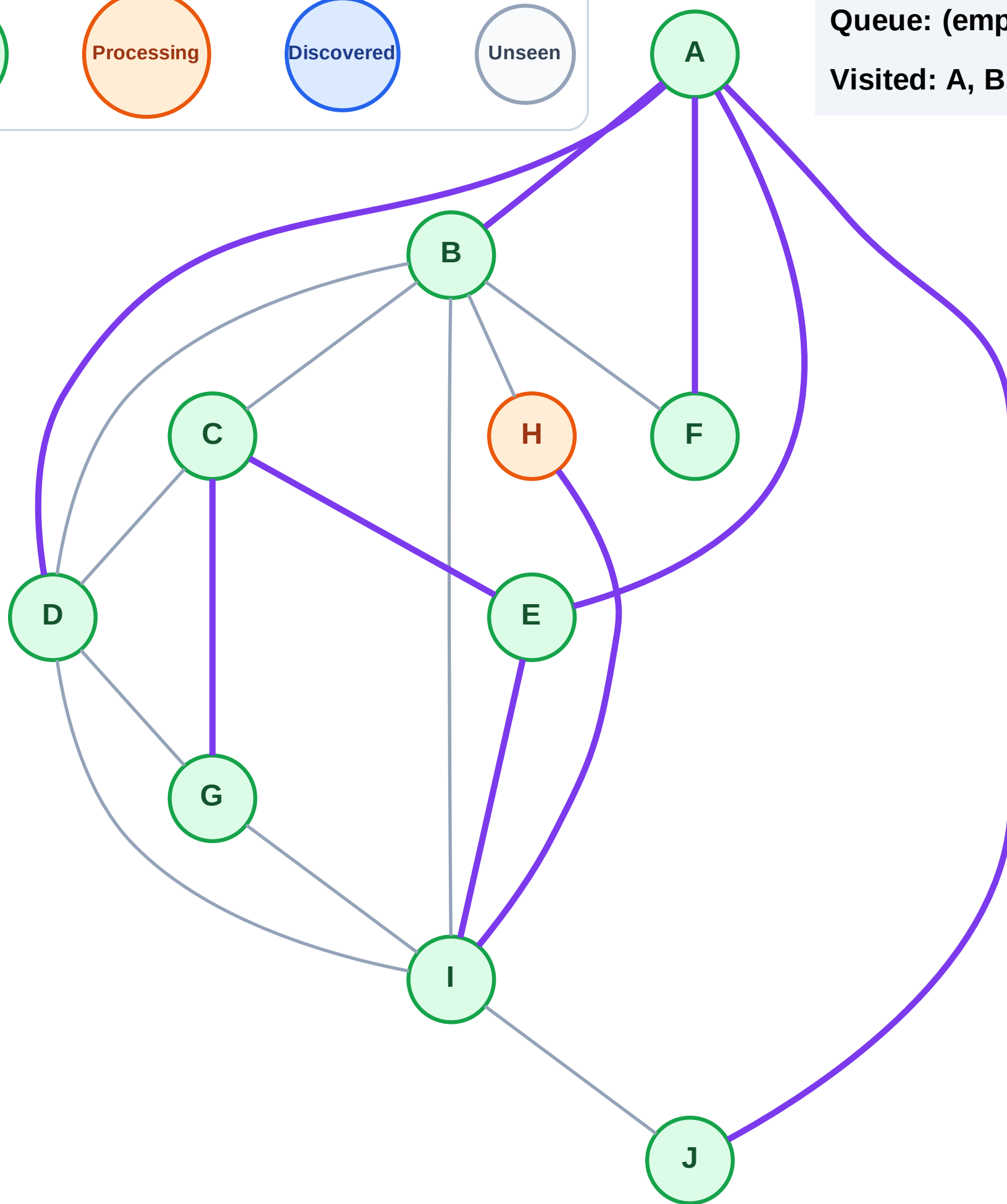
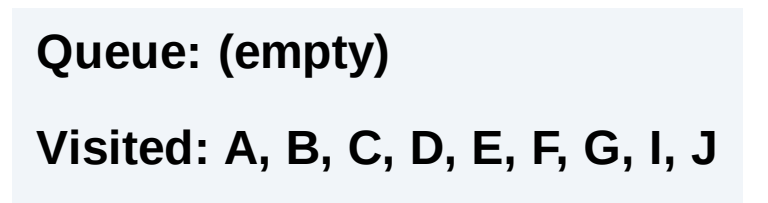
Queue: H

Visited: A, B, C, D, E, F, G, I, J



Step 20: Process node H

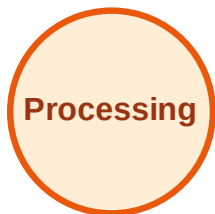
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BFS Traversal

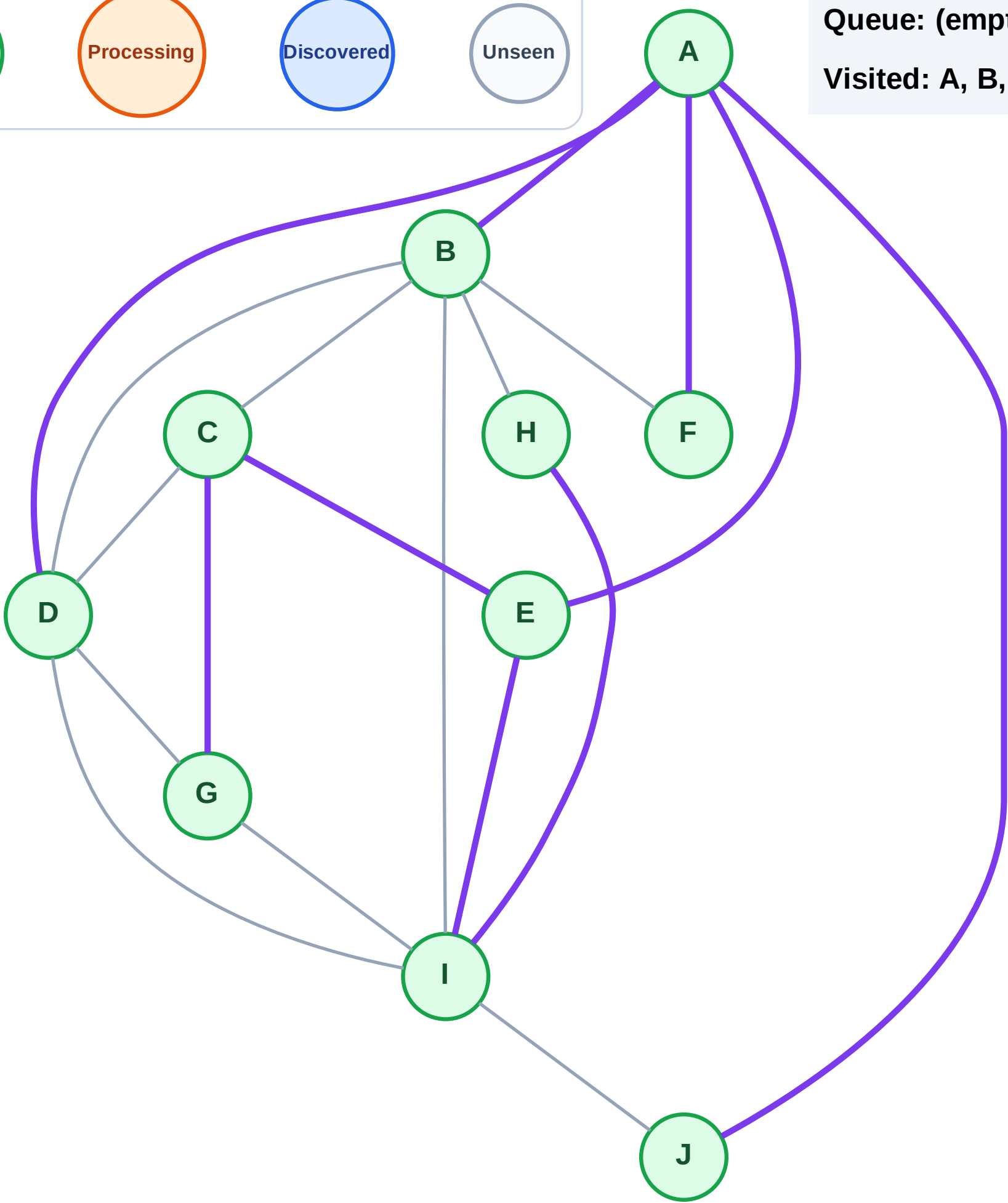
Step 21: No new neighbors from H

Legend



Queue: (empty)

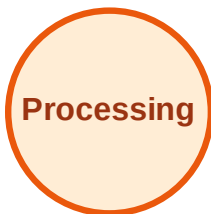
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

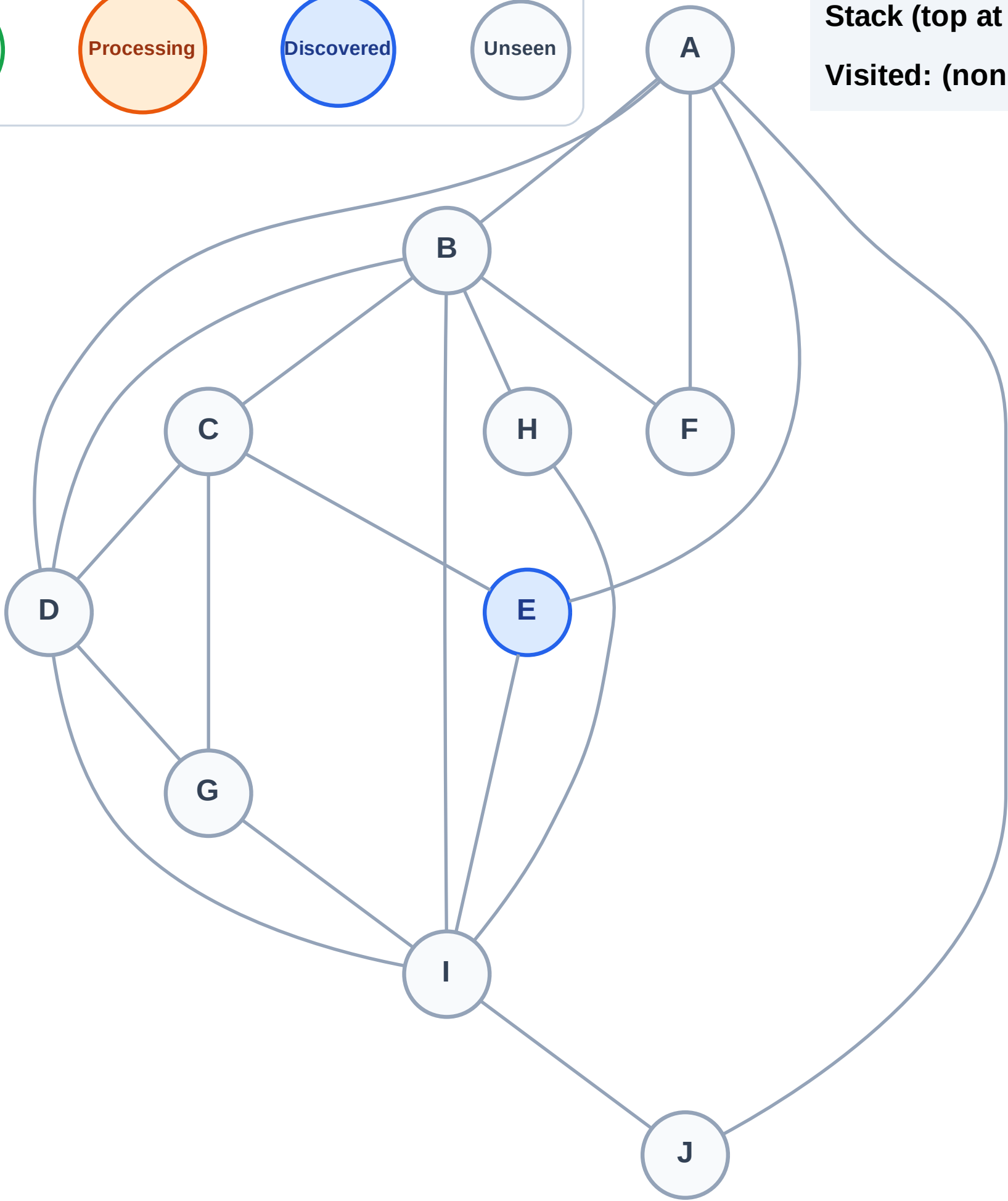
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

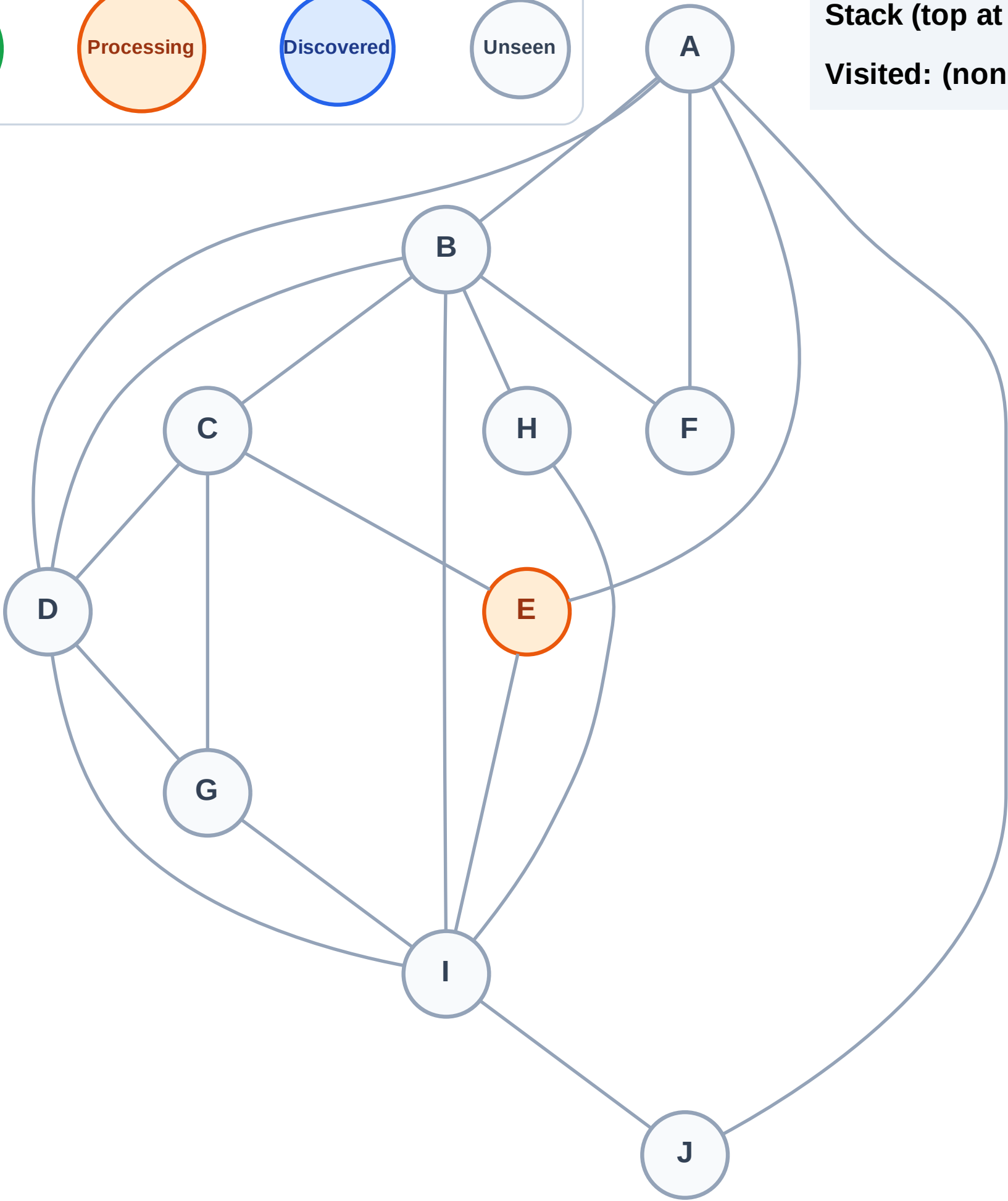
Complete

Processing

Discovered

Unseen

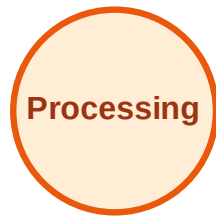
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

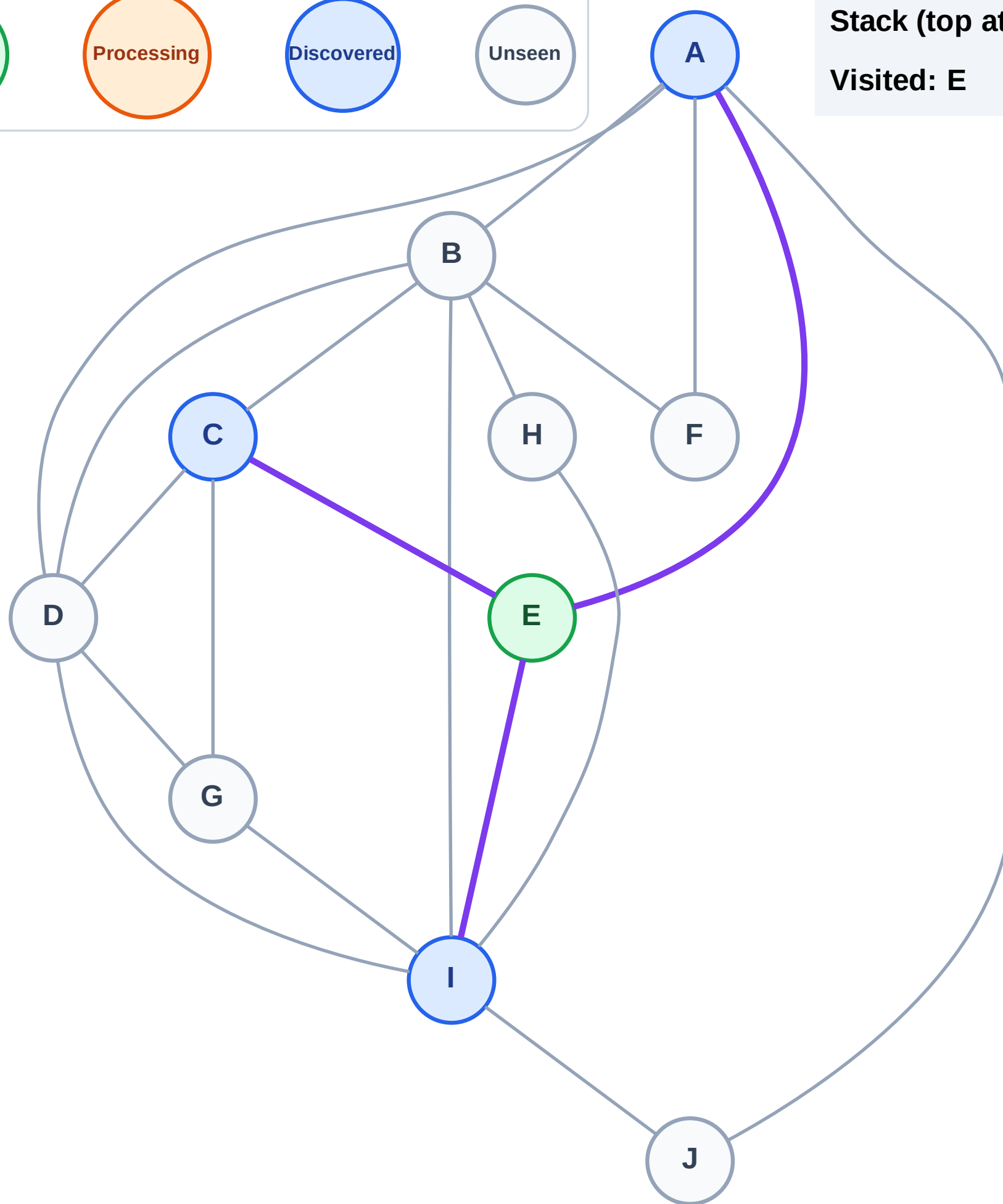
Step 3: Discover I, C, A from E

Legend



Stack (top at right): I | C | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

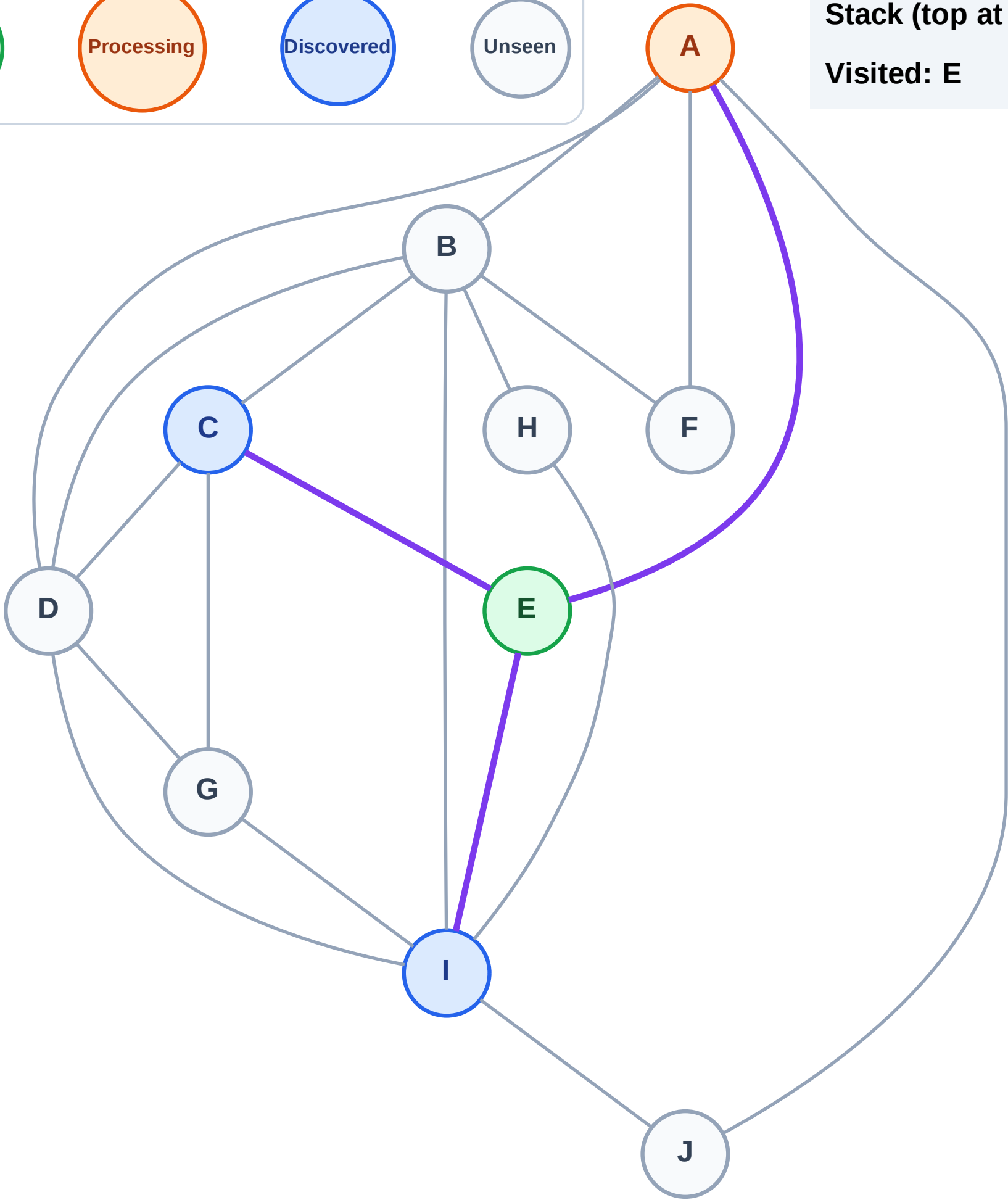
Processing

Discovered

Unseen

Stack (top at right): I | C

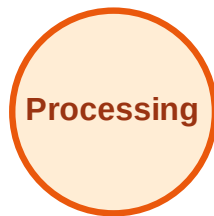
Visited: E



DFS Traversal

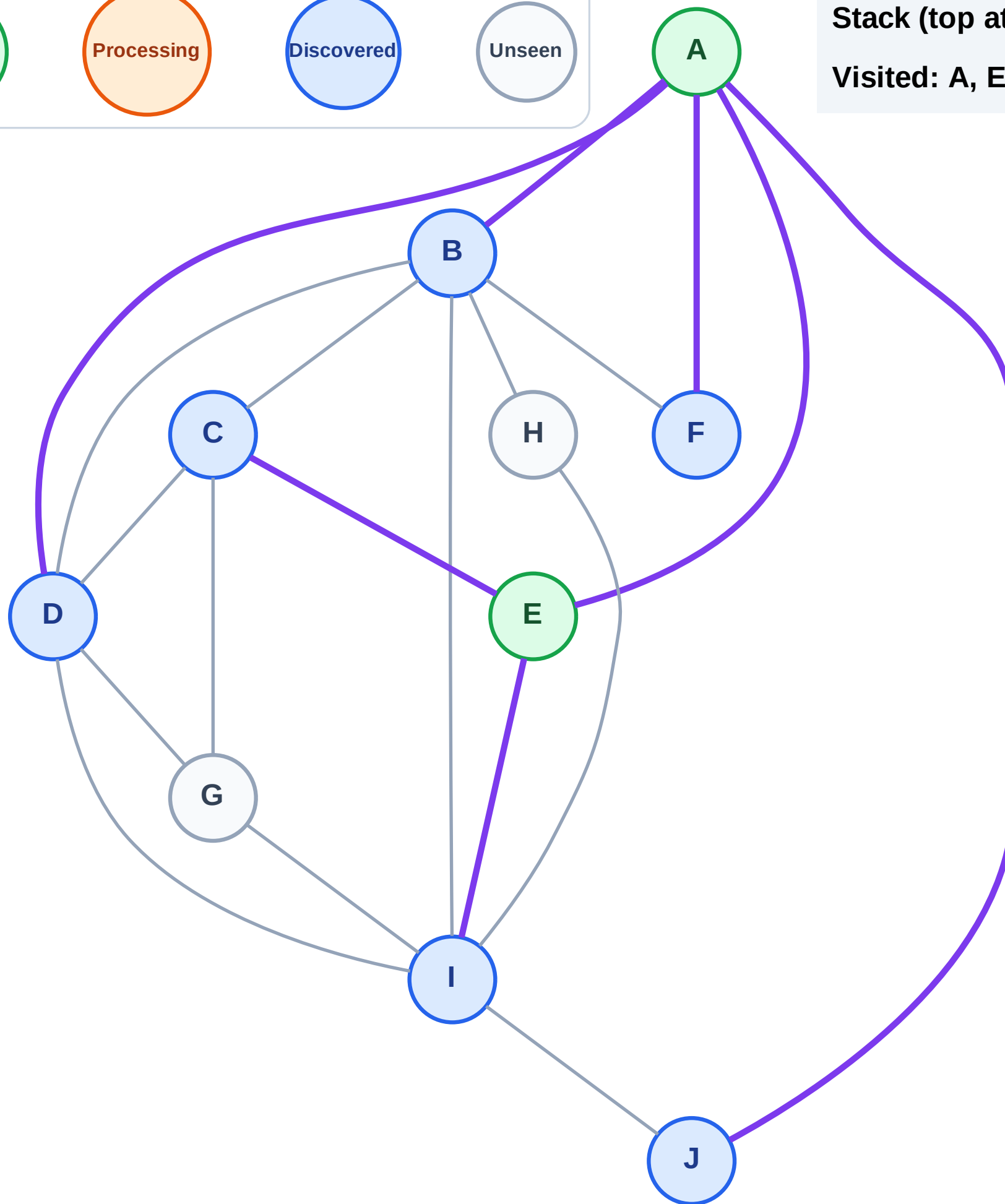
Step 5: Discover J, F, D, B from A

Legend



Stack (top at right): I | C | J | F | D | B

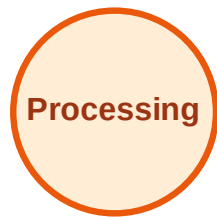
Visited: A, E



DFS Traversal

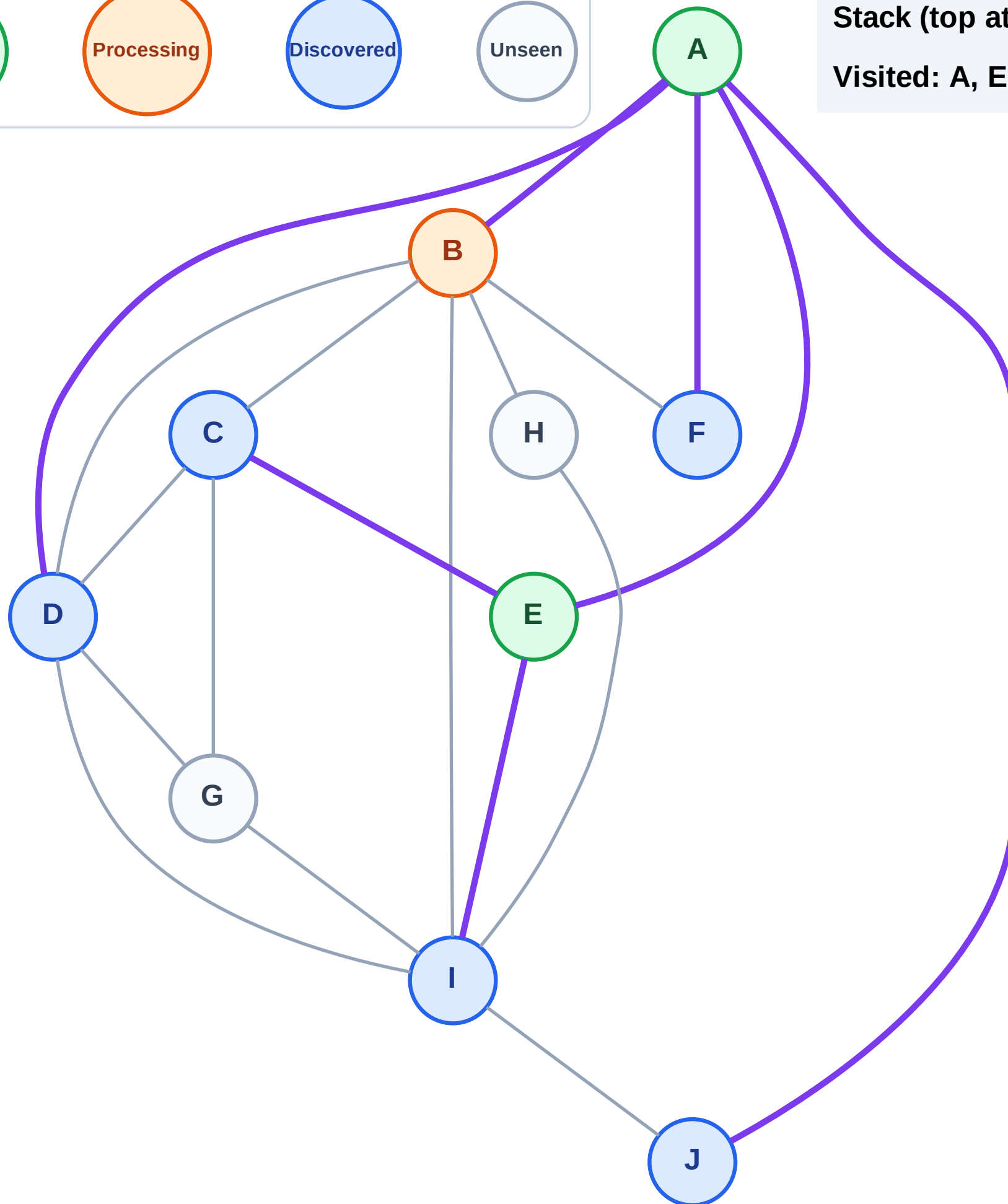
Step 6: Process node B

Legend



Stack (top at right): I | C | J | F | D

Visited: A, E



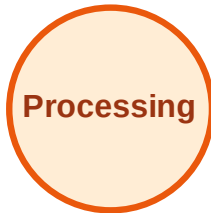
DFS Traversal

Step 7: Discover H from B

Legend



Complete



Processing



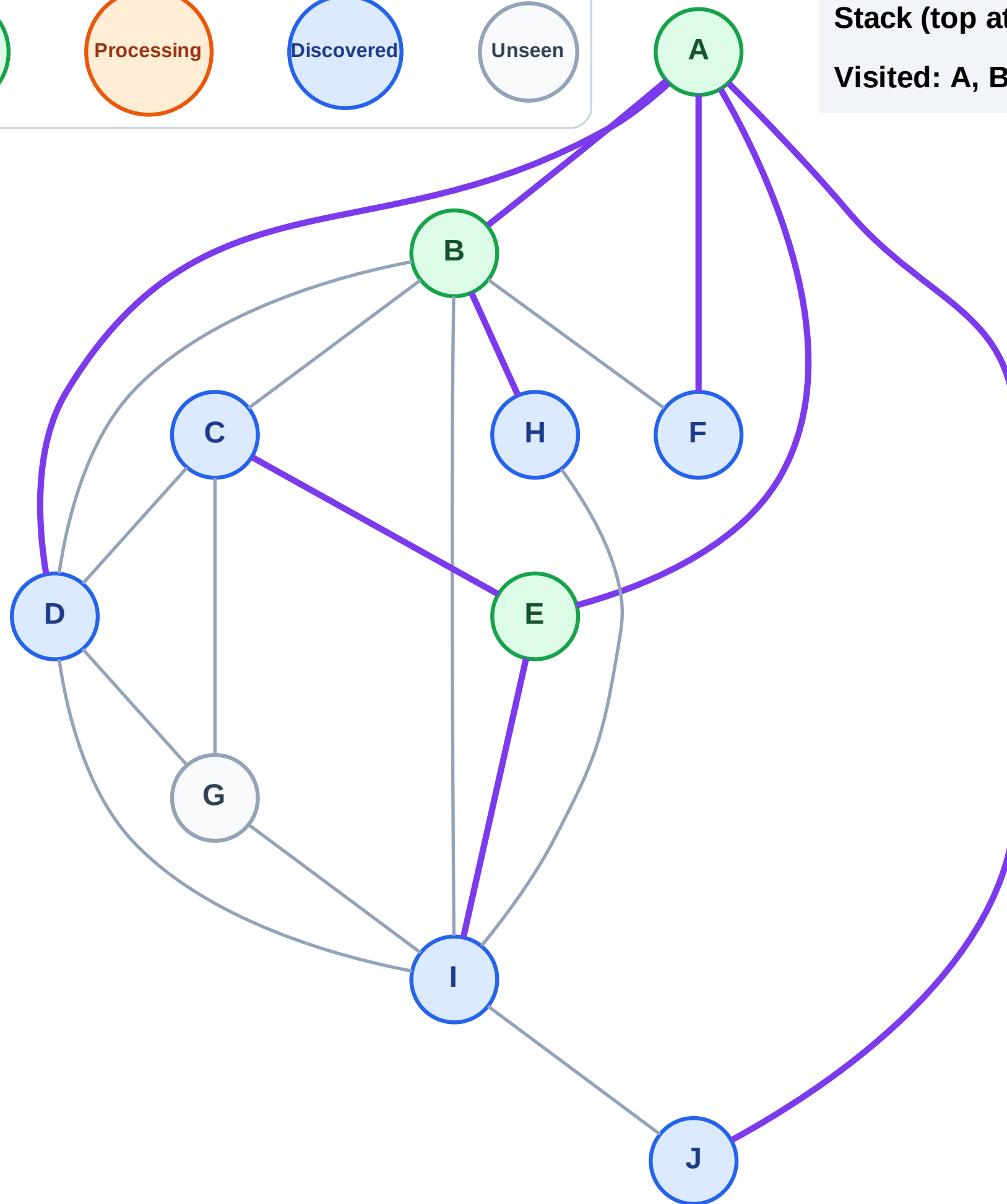
Discovered



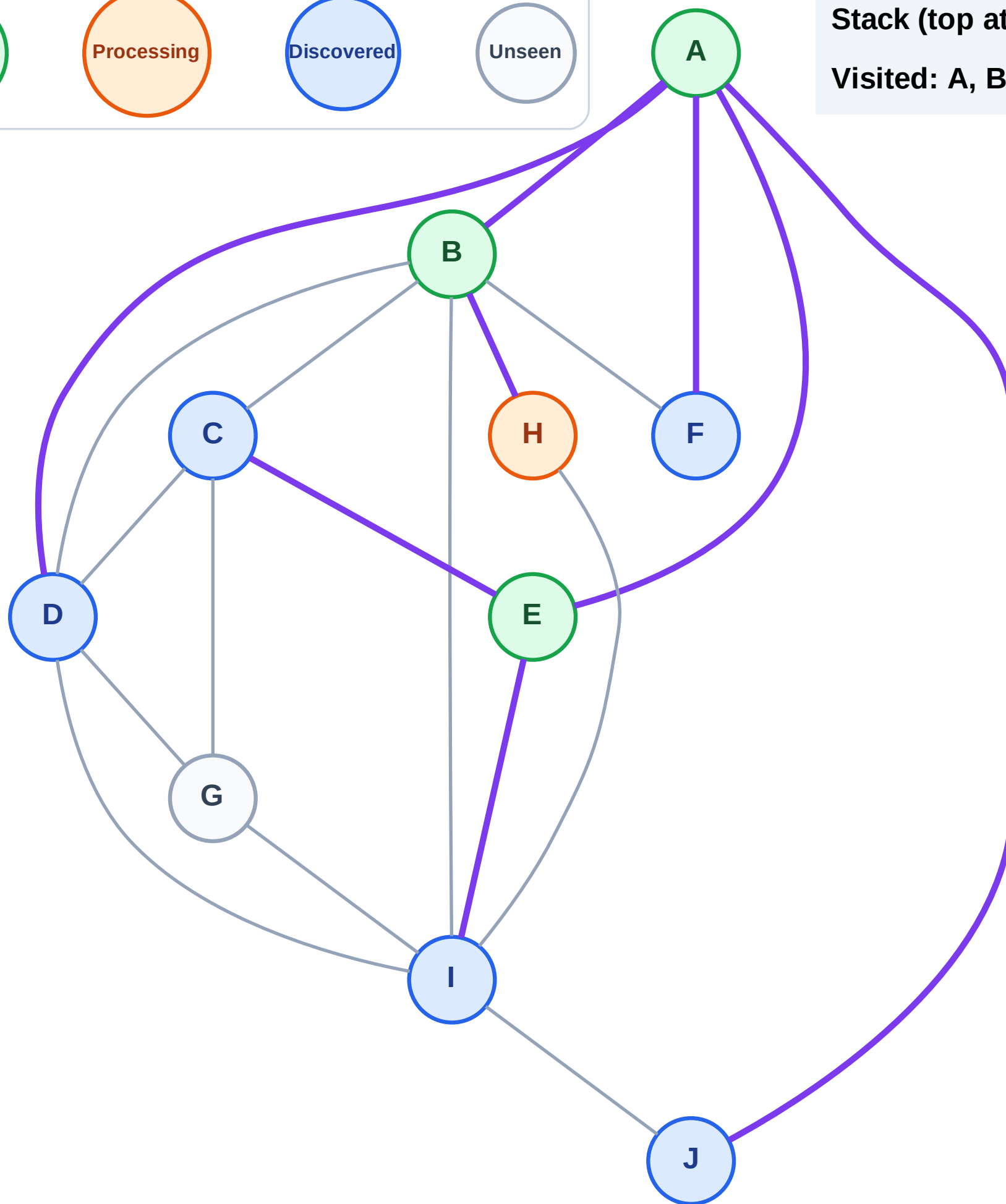
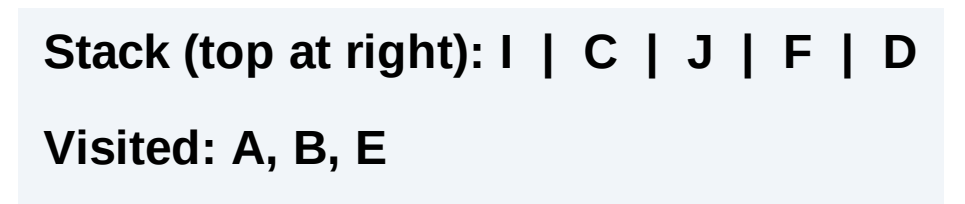
Unseen

Stack (top at right): I | C | J | F | D | H

Visited: A, B, E



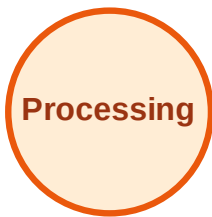
Step 8: Process node H



DFS Traversal

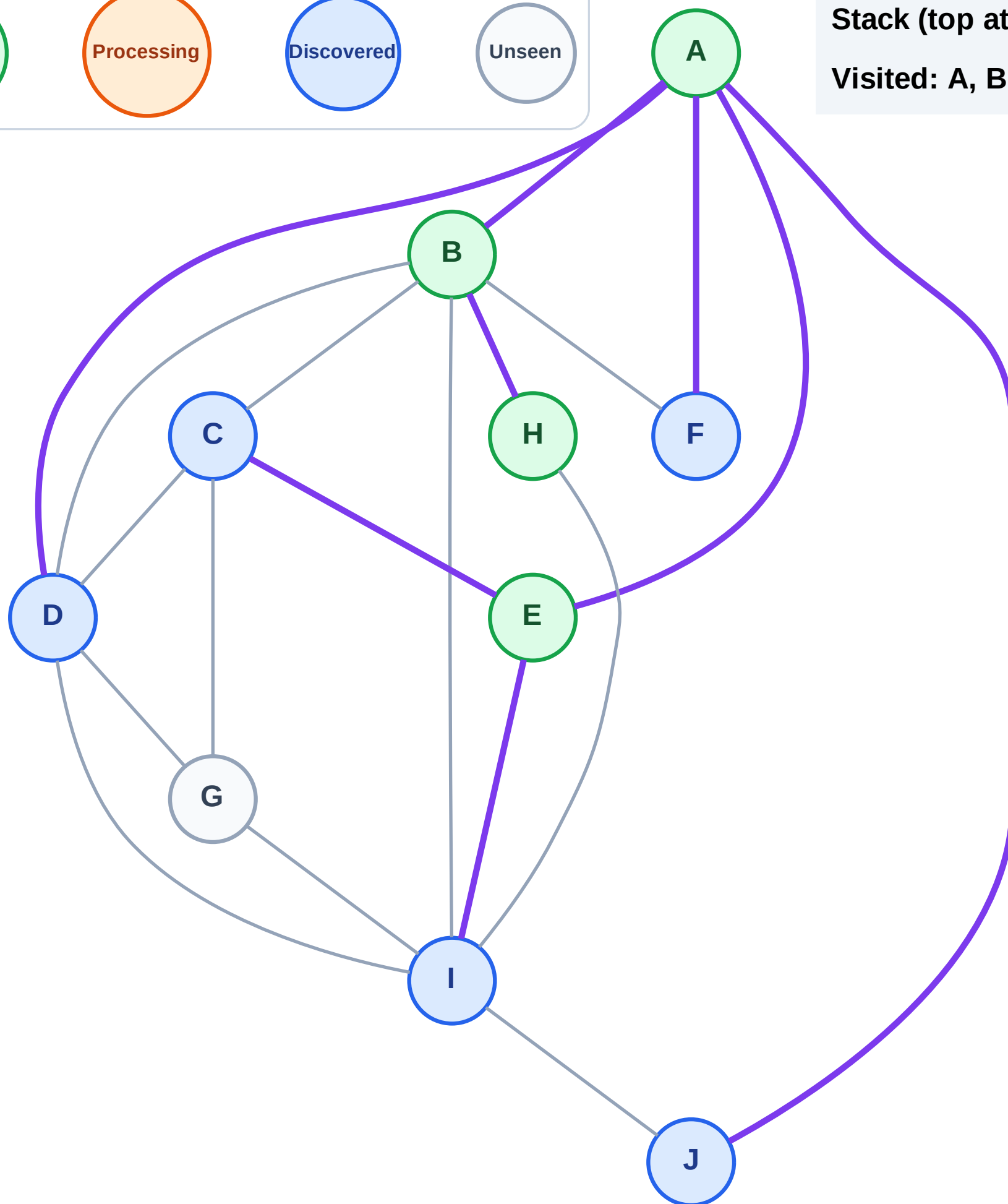
Step 9: No new neighbors from H

Legend



Stack (top at right): I | C | J | F | D

Visited: A, B, E, H



DFS Traversal

Step 10: Process node D

Legend



Complete



Processing



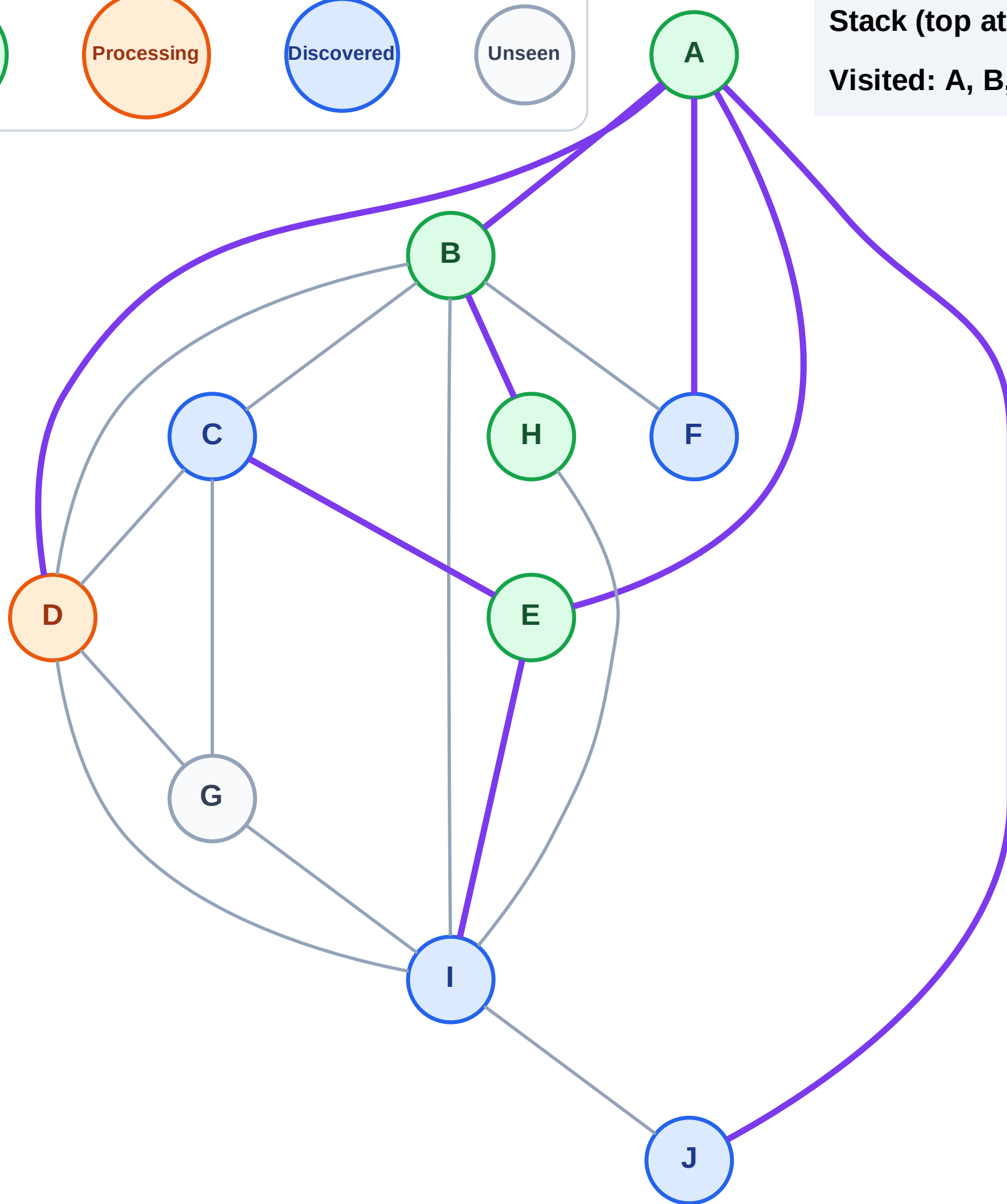
Discovered



Unseen

Stack (top at right): I | C | J | F

Visited: A, B, E, H



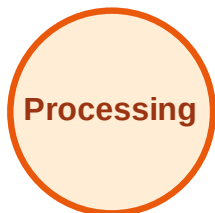
DFS Traversal

Step 11: Discover G from D

Legend



Complete



Processing



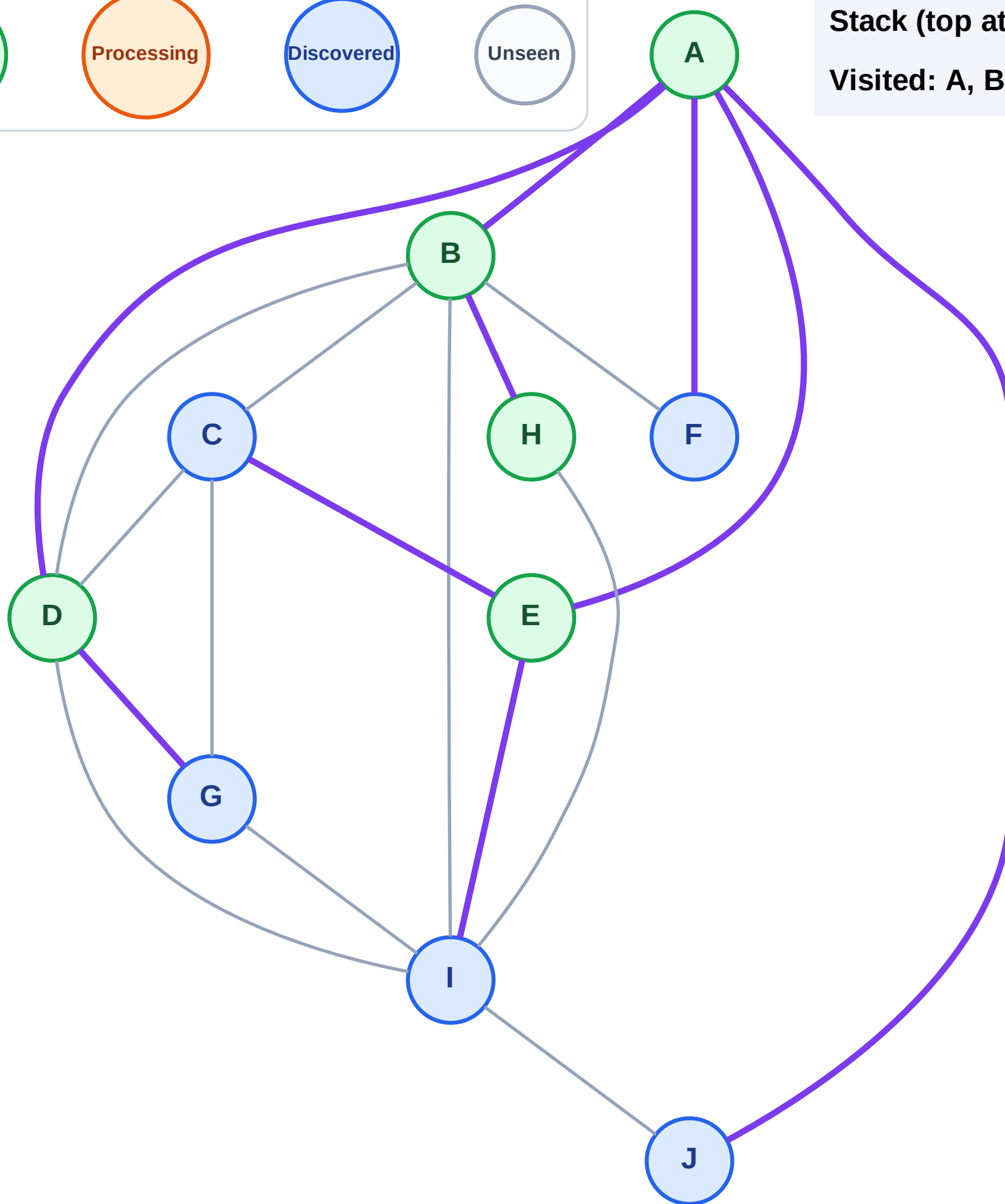
Discovered



Unseen

Stack (top at right): I | C | J | F | G

Visited: A, B, D, E, H



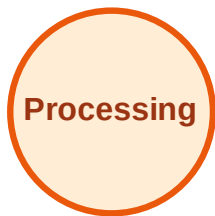
DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



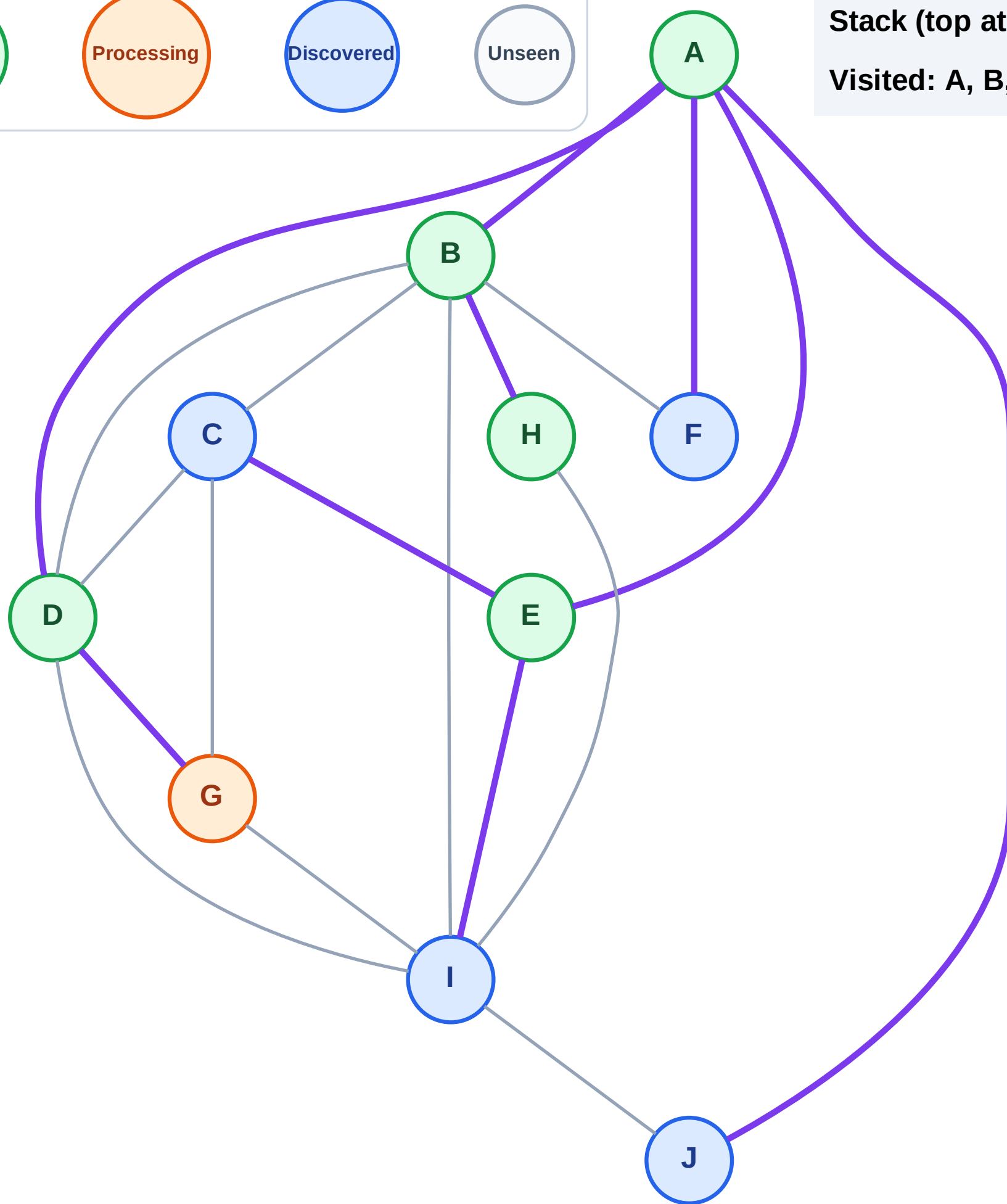
Discovered



Unseen

Stack (top at right): I | C | J | F

Visited: A, B, D, E, H



DFS Traversal

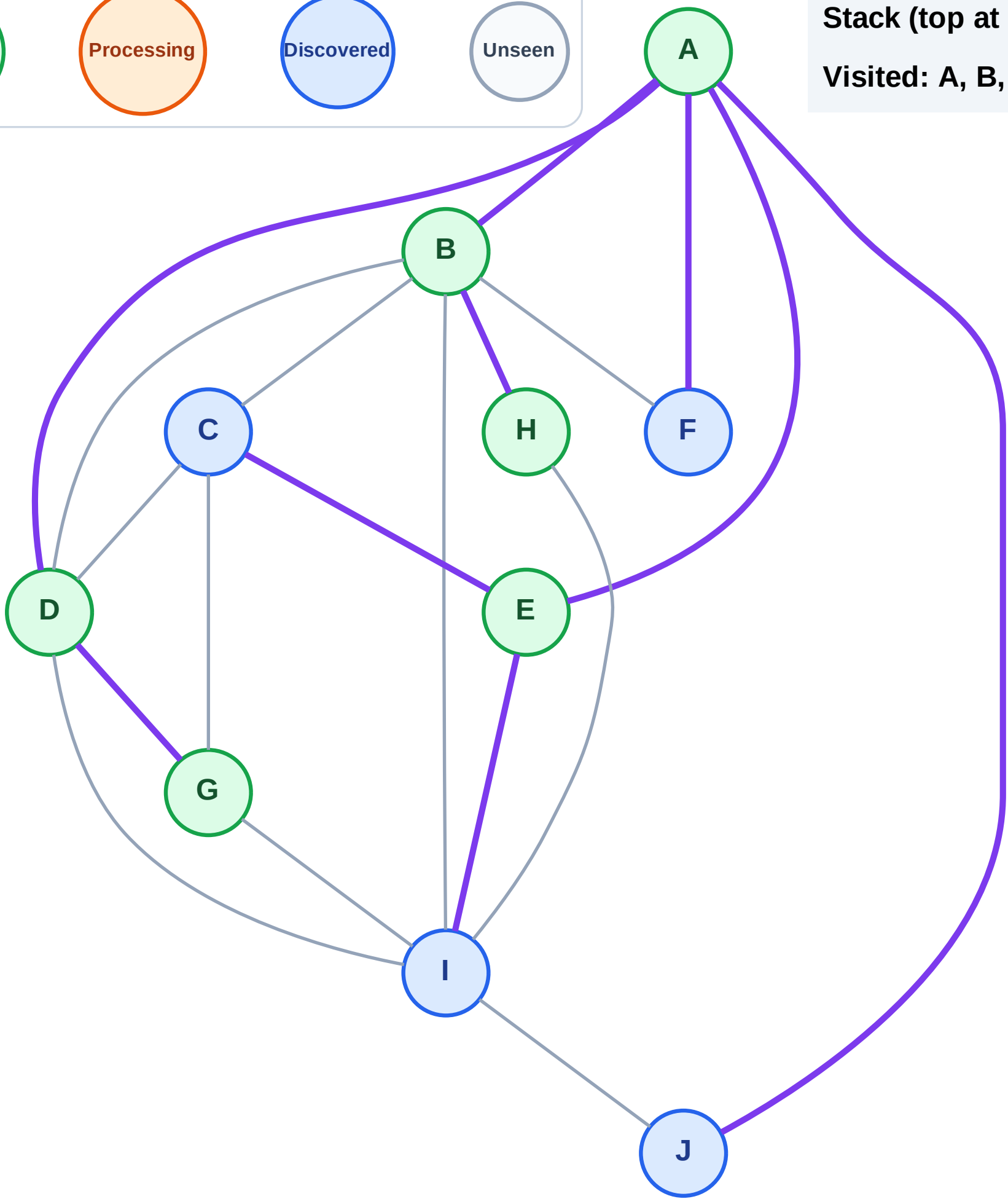
Step 13: No new neighbors from G

Legend



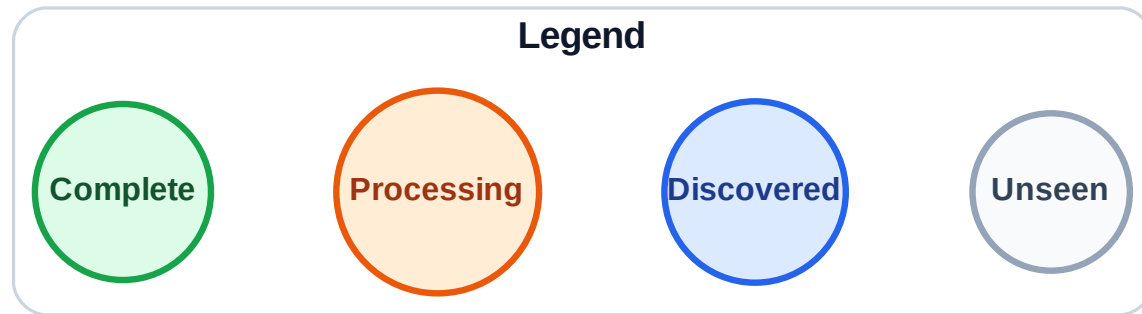
Stack (top at right): I | C | J | F

Visited: A, B, D, E, G, H



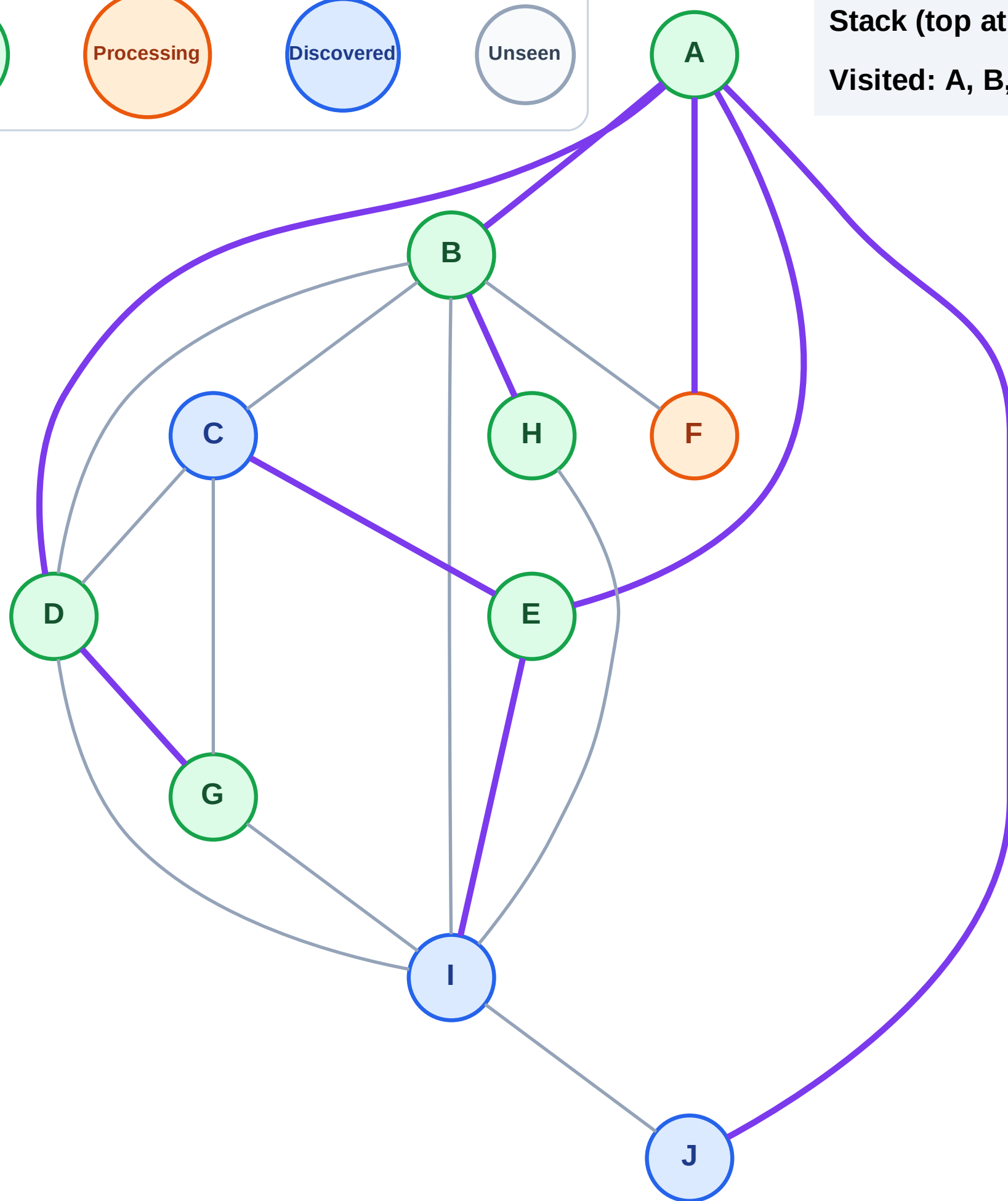
Step 14: Process node F

Step 14: Process node F



Stack (top at right): I | C | J

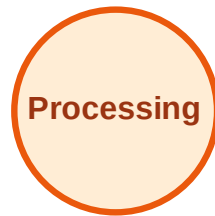
Visited: A, B, D, E, G, H



DFS Traversal

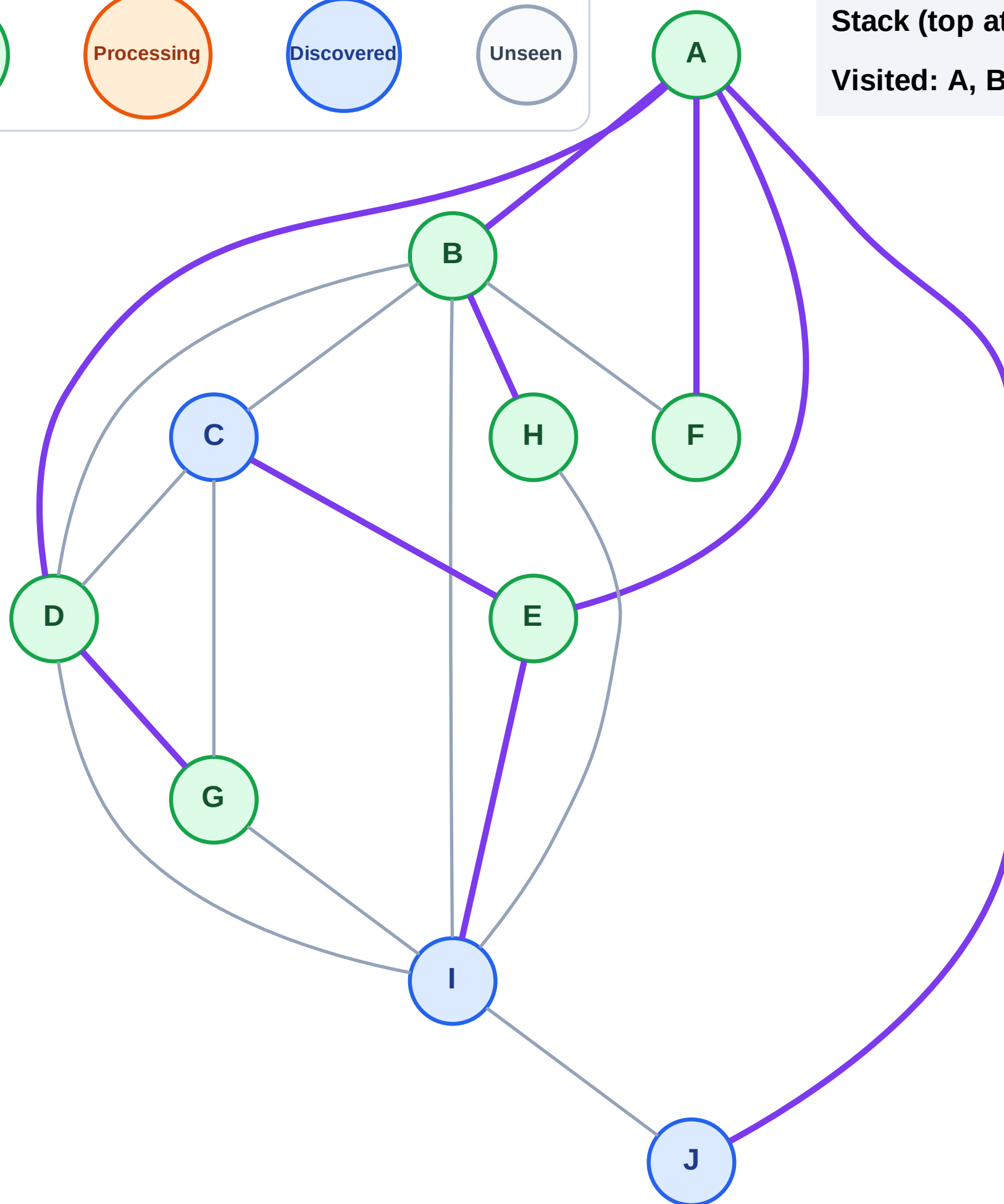
Step 15: No new neighbors from F

Legend



Stack (top at right): I | C | J

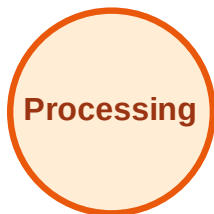
Visited: A, B, D, E, F, G, H



DFS Traversal

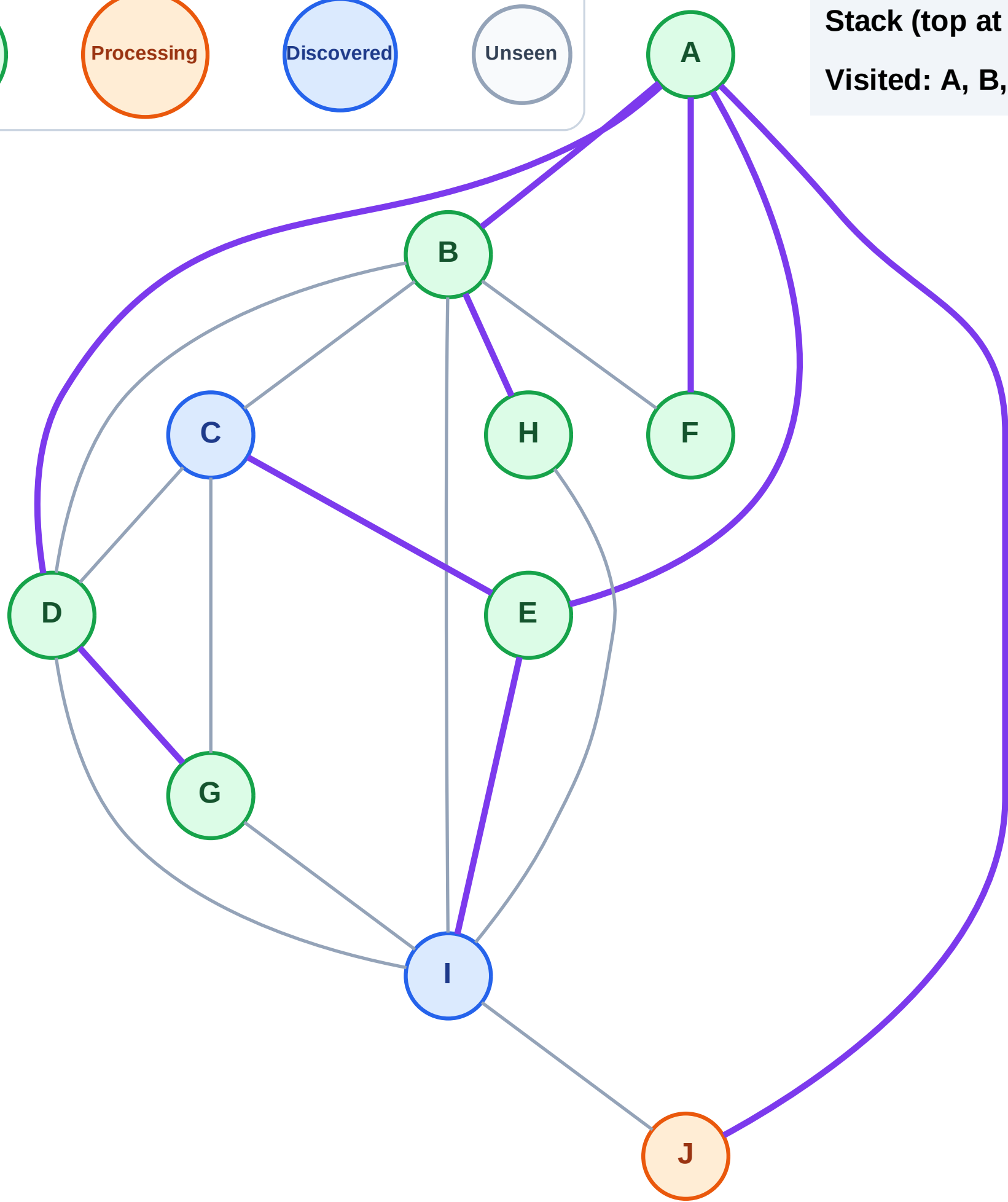
Step 16: Process node J

Legend



Stack (top at right): I | C

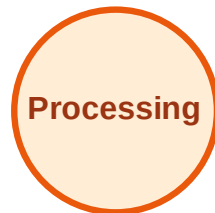
Visited: A, B, D, E, F, G, H



DFS Traversal

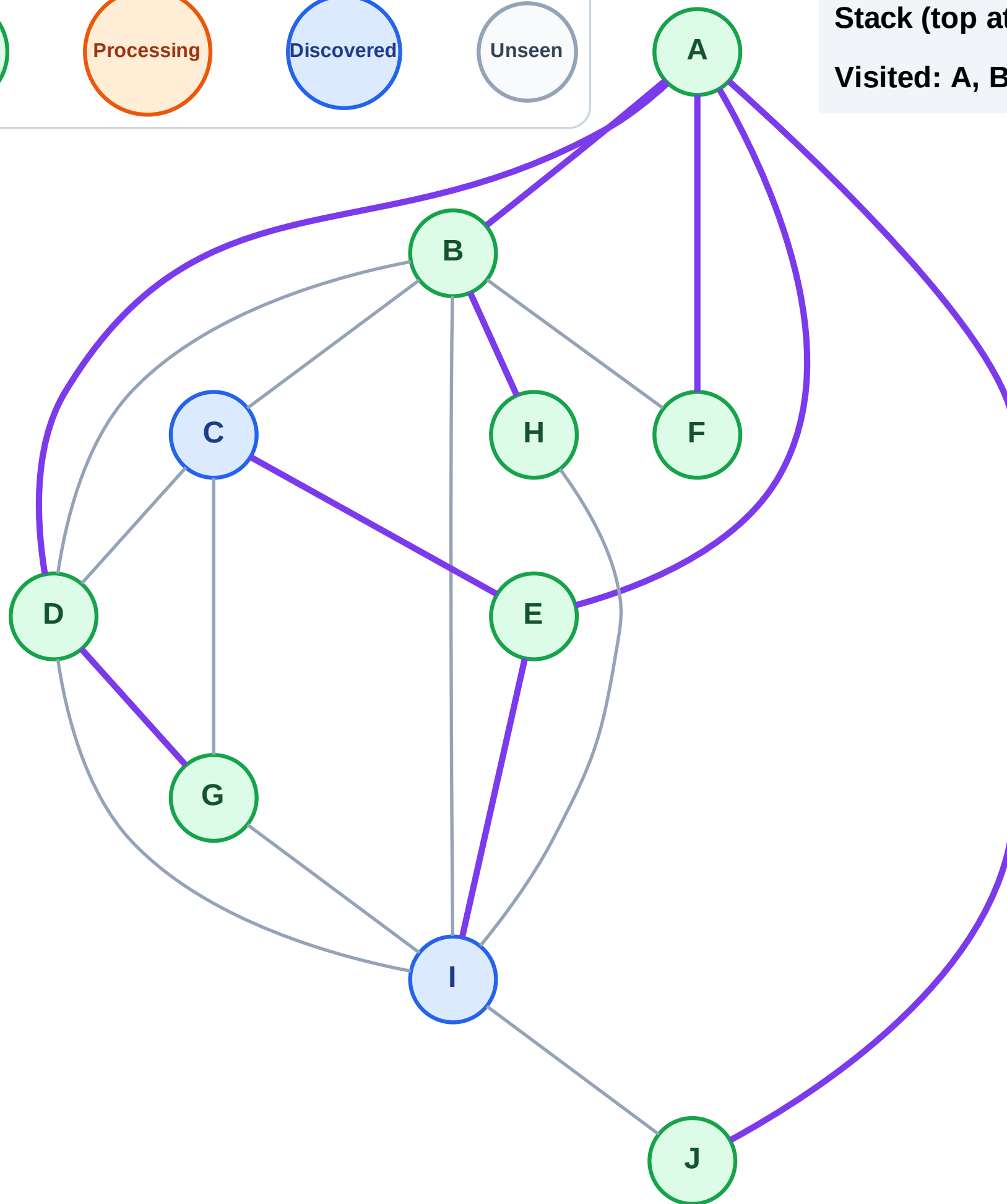
Step 17: No new neighbors from J

Legend



Stack (top at right): I | C

Visited: A, B, D, E, F, G, H, J



DFS Traversal

Step 18: Process node C

Legend

Complete

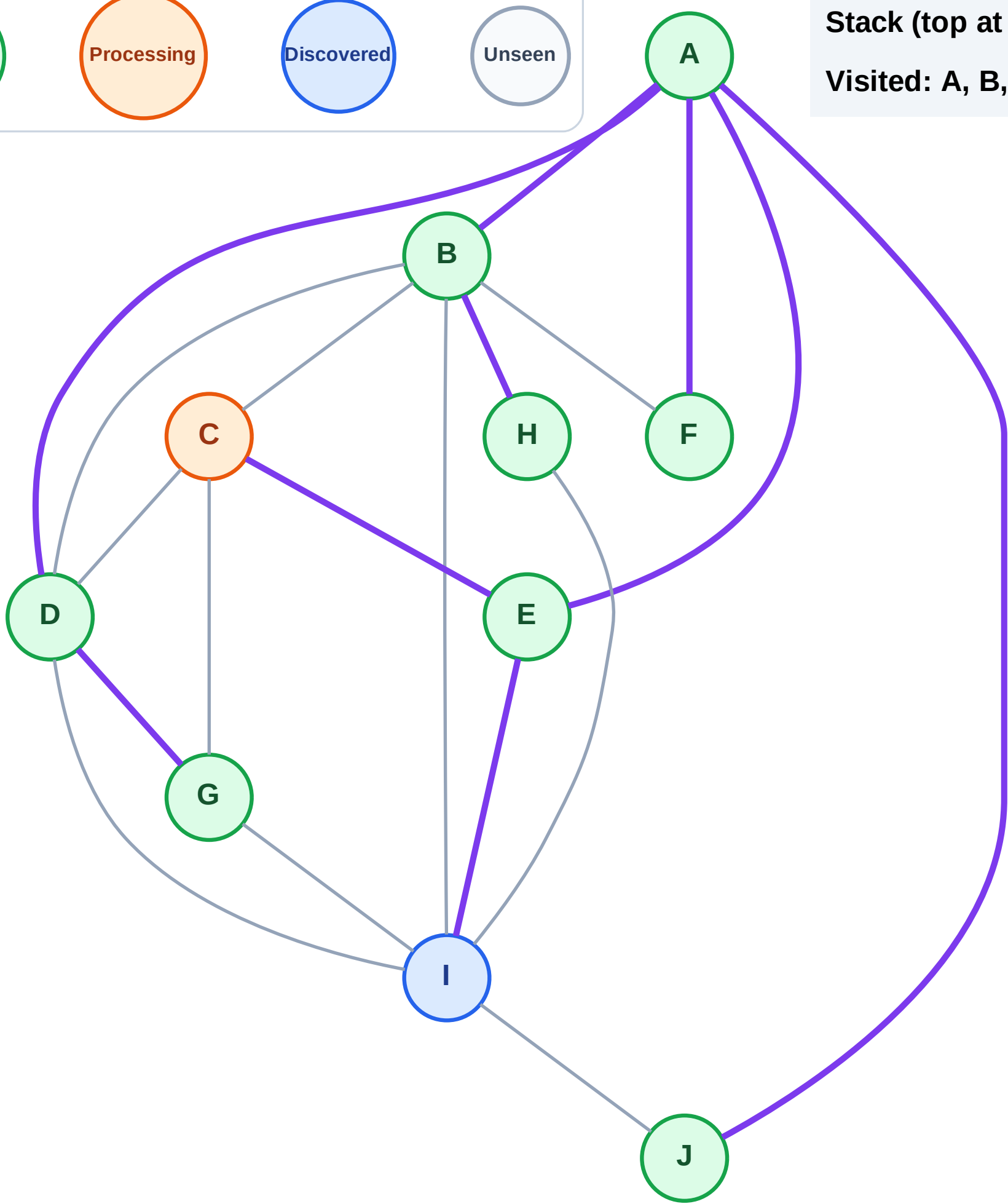
Processing

Discovered

Unseen

Stack (top at right): I

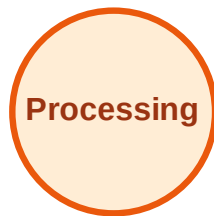
Visited: A, B, D, E, F, G, H, J



DFS Traversal

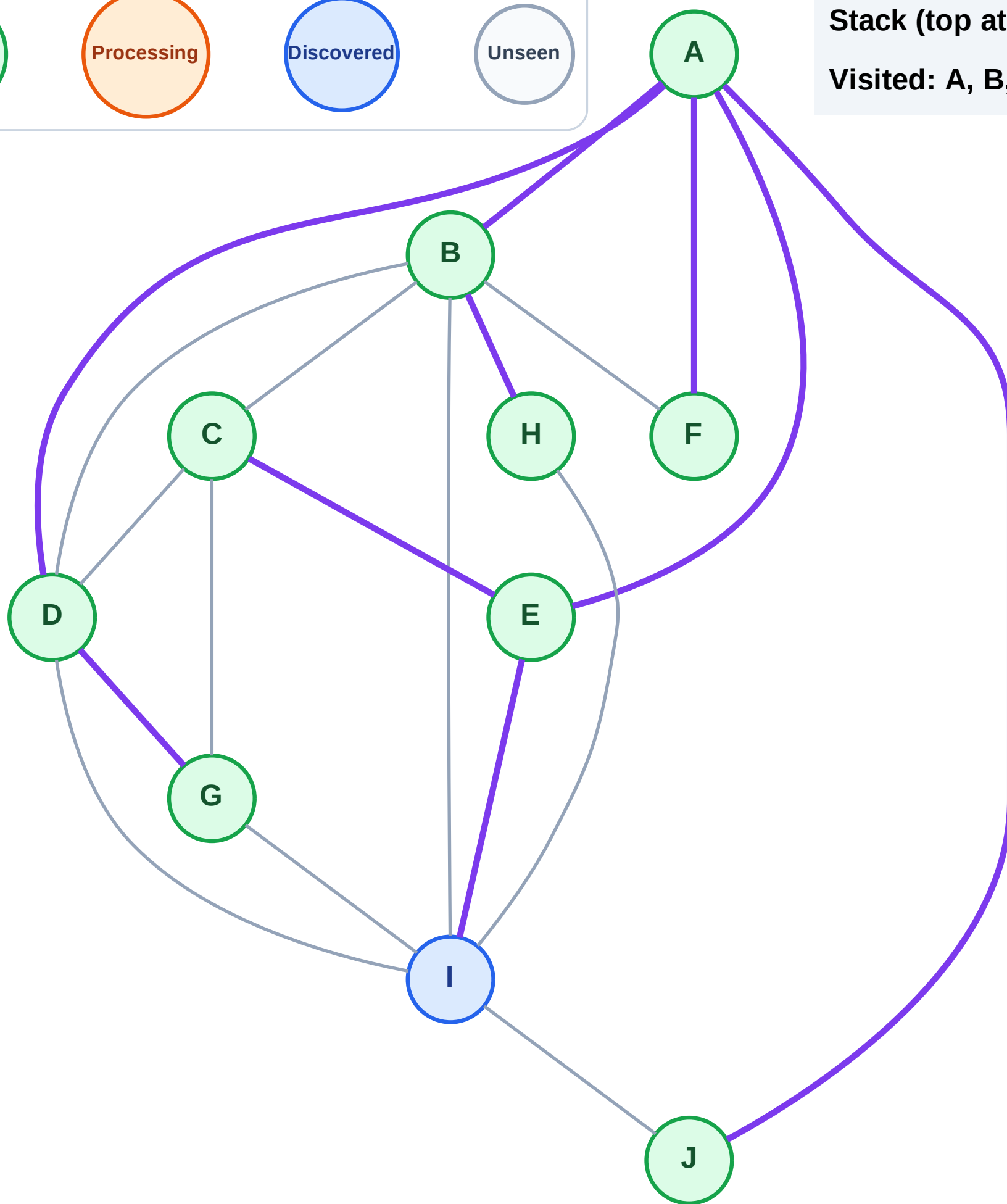
Step 19: No new neighbors from C

Legend



Stack (top at right): I

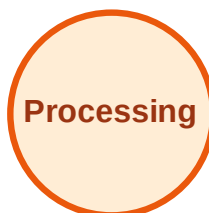
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

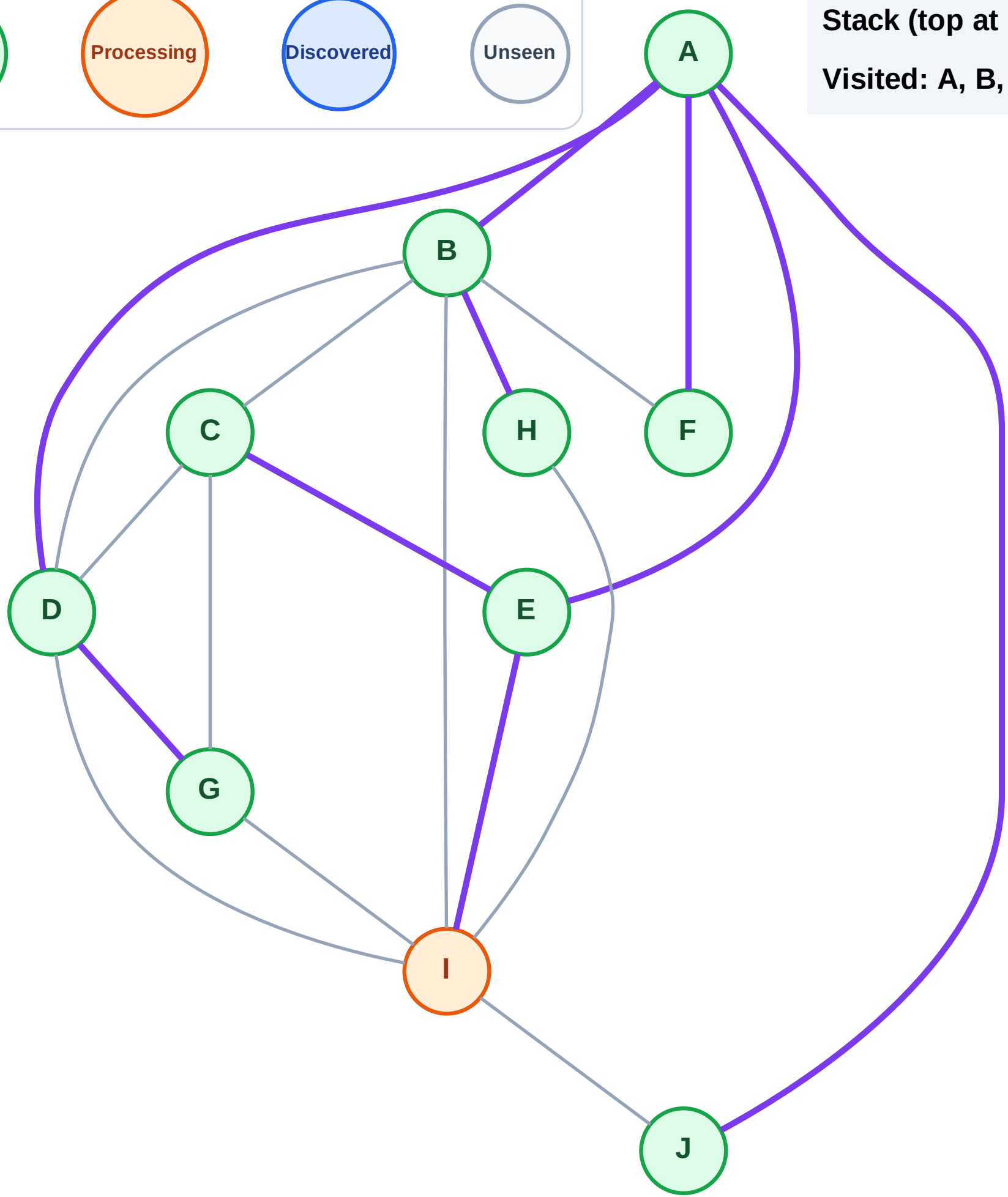
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

